



Enhancing university education by bringing industry into the classroom

EVENT HANDBOOK

2025 FORMULA SAE-A

11-14 December 2025

**Calder Park Raceway
377 Calder Freeway
Calder Park, Victoria**

WORDS FROM THE SAE-A PRESIDENT

Martha Oplopiadis

President, SAE-Australasia



On behalf of the SAE-A, I welcome you to the Calder Park Raceway for the 2025 Formula SAE-A.

Once again, teams from all around the world will turn theory into practice as they apply their STEM learnings to a real-world competitive situation and vie for top honours across a range of events.

The Board of SAE-A and the Organising Committee are proud to host the Formula SAE-A. The event reflects our commitment to ensuring University students are offered the most innovative, dynamic and practical experiences possible to ensure they finish their education job-ready and able to embark on their exciting new careers.

Whether you are part of an international or Australian team, welcome, and I encourage you to make the most of your visit opportunities through this amazing program.

I will personally be attending as a volunteer and I look forward to meeting as many of you as possible, to hear your stories and see first-hand the enthusiasm and passion that defines Formula SAE-A.

Whatever your role in the 2025 Formula SAE-A, I hope you enjoy your time on and off the track.

EVENT PARTNERS



SUPPORTERS



Table of Contents

EVENT HANDBOOK	1
2025 FORMULA SAE-A	1
WORDS FROM THE SAE-A PRESIDENT	2
EVENT PARTNERS	3
TEAM REQUIREMENTS	6
GENERAL EVENT INFORMATION	7
TEAMS & PIT GARAGES 2025	8
ORGANISING COMMITTEE	8
TRAVEL & ACCOMMODATION	10
SITE LAYOUT	11
ONSITE WORKSHOP FACILITIES	12
SAFE WORK PROCEDURE	13
ELECTRIC VEHICLE SAFETY – OVERALL	14
ELECTRIC VEHICLE SAFETY- ONSITE	15
ONSITE ACCUMULATOR CLEAN ROOM	16
FORMULA SAE-A STUDENT (FS) TEAMS	19
EVENT SCHEDULE	20
STATIC EVENTS & TECHNICAL INSPECTION SCHEDULE	21
AWARDS CEREMONY	22
VEHICLE INSPECTION PANEL	23
TECHNICAL INSPECTION	24
TECHNICAL INSPECTION – CONT	25
WEIGH STATION & TILT TABLE	26
FUEL STATION	26
BRAKE AND NOISE TEST	27
RAIN TEST	28
ACCUMULATOR INSPECTION	28
DESIGN EVENT	29
COST EVENT	30
COST EVENT (continued)	31

PRESENTATION EVENT	32
DYNAMIC EVENTS OVERVIEW	34
DYNAMIC EVENTS OVERVIEW	35
PRACTICE TRACK	36
ACCELERATION EVENT	36
SKID PAD EVENT	36
AUTONOMUS VEHICLE DEMONSTRATION	38
AUTOCROSS EVENT	39
ENDURANCE & EFFICIENCY EVENTS	40
PARC FERME	41
PARC FERME (continued)	42
COFFEES	43
LOCAL SHOPPING	43
ONSITE CATERING	43
EQUIPMENT SUPPLIERS	44
VEHICLE SHIPPING	45
VEHICLE SHIPPING (continued)	46
VOLUNTEER INFORMATION	47
VOLUNTEER INFORMATION (continued)	48
VOLUNTEER INFORMATION (continued)	49
VOLUNTEER INFORMATION (continued)	50

TEAM REQUIREMENTS

The Team Leader and Faculty Advisor or (OH&S representative), if not the Faculty Advisor, must visit the Administration Office to complete registration upon arrival. The following documents must be presented. The templates can be found online <https://www.saea.com.au/rules-documents-templates>

- A **completed** Self-assessed Technical Inspection form signed by the Faculty Advisor.
- Copy of the list of Egress Times.
- Confirmation of EV Compliance and Safety Protocols.
- Declaration of Hazardous Materials book.
- Cost Report addenda covering any changes to the car (if relevant). Must be submitted by 9.00am on December 11th.
- Payment for any outstanding student registrations.
- Teams are reminded they must also bring a copy of all rules questions they have submitted and related responses.

ONSITE REGISTRATION

Teams will be provided with the following equipment:

- Energy meters will be issued to the EV team at registration.
- Thermal strips will be fitted to accumulators at EV Accumulator Inspection.
- An AV data logger will be issued at Autonomous Technical Inspection.
- Timing transponder before track events.

MOTORSPORT AUSTRALIA

The event is underwritten by Motorsport Australia, so refer to their website for the latest information regarding the level of coverage. <https://sport.ajg.com.au/motorsport-australia/>

Please note that team members and vehicles, if off-site from the Calder Park Raceway, are not covered by the event insurance policy and, as such, are at your own risk

MEDICAL TREATMENT

First Aid will be available onsite from Thursday to Sunday. For medical treatment outside event hours, please visit a local medical provider. Incidents and near-misses must be reported to the Administration Office.

HOSPITALS

The Royal Melbourne Hospital – Grattan Street, Parkville Tel: (03) 9342 7000
Western Health Hospital Footscray – 160 Gordon Street, Footscray Tel: (03) 8345 6666

Call “000” in an emergency
Police Assistance Non-emergency 131 444

AFTER HOUR CONTACTS

The following numbers can be called outside event hours:

Adrian Feeney	+61 432 689 114
Angela Krepcik	+61 408 218 158
Rose De Amicis	+61 403 267 166

GENERAL EVENT INFORMATION

SITE ACCESS: The event site can be accessed from the main gate on Calder Freeway from 7:00 am each day.

SAFETY & REPORTING INCIDENTS: Formula SAE-Australasia (FSAE-A) is conducted under a Motorsport Australia (MA) event permit. The MA OH&S policy can be viewed online (motorsport.org.au) and at the Administration Office. Participants are involved in the event by choice and must bear responsibility for their own behaviour, as well as those around them. Any incidents (including near misses) must be reported to the Administration Office.

EVENT PARTICIPATION: All participants must be

- at least 18 years old or provide a signed declaration by their parent or guardian
- be affiliated with a university
- be a member of SAE-A or their equivalent local association.

DOCUMENTS & FORMS: All documents/forms can be obtained online at <https://www.saea.com.au/rules-documents-templates>

VEHICLE CLASSES: Vehicles will compete in the Electric Vehicle (EV) or Internal Combustion (IC) class. Both classes compete in the same events but are independently scored/judged. The AV class is a demonstration only and judged separately.

LICENSING AND INSURANCE: Drivers must obtain personal medical insurance, as well as an MA Speed License (Formerly Level 2S License). All participants must submit a signed MA disclaimer form.

BEHAVIOUR: Alcohol, drugs, and animals are not permitted onsite. Drones, motorcycles, bicycles, skateboards, rollerblades, scooters, and similar person-carrying devices are also not permitted.

FOOTWEAR: Participants and spectators must wear closed-toe shoes. Shoes with rubber soles are mandatory for all EV team members. Persons not complying with these requirements may be asked to leave the venue.

SECURITY: Please keep your equipment secure at all times, particularly tools, phones and laptop computers.

PHOTOGRAPHERS: The event organisers may authorise professional photographers to enter dynamic areas. Photographers must be registered, sign a waiver, wear high visibility clothing, and be accompanied by a spotter.

FIRE EXTINGUISHERS: Fire extinguishers must be readily accessible, and team members must be competent in their use.

POWER TOOLS: Power tools connected to the mains electricity supply and/or used for the removal of material (e.g., steel or aluminium) can only be used in designated workshops. Small battery-operated hand tools used for assembly/disassembly may be used in pit garages, provided they do not produce excessive noise. Any electrical equipment must be tested and tagged before coming onsite.

OPEN FIRES & SMOKING: BBQs NOT ALLOWED. Smoking is only permitted in designated areas.

RUBBISH, WASTE FUEL & OIL: Please dispose of all rubbish in the waste bins provided. No rubbish may be left behind after the event. **Please remember to care for the environment by placing rubbish in the correct bins.**

Do not mix waste with recyclables

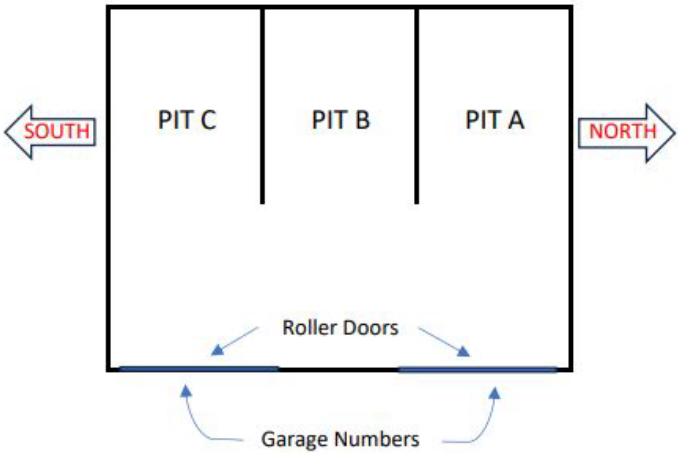
Waste fuel or oil must be taken to the Fuel Station for disposal.

TEAMS & PIT GARAGES 2025

Team Name	Vehicle Number	Garage	Bay
University of Wollongong	E85	5/6	A
Swinburne University of Technology	E17	5/6	B
The University of Melbourne	E68	5/6	C
Tokyo Denki University	21	7/8	A
The University of Sydney	E22	7/8	B
UNSW ADFA	E15	7/8	C
Edith Cowan University	7	9/10	A
Institut Teknologi Sepuluh Nopember	E74	9/10	B
University of Canterbury	E13	9/10	C
University of Newcastle	E3	11/12	A
The University of Queensland	E42	11/12	B
The University of Queensland	A48	11/12	C
Australian National University	61	13/14	A
Griffith University	111	13/14	B
The University of Adelaide	E8	13/14	C
Alfaisal University	70	15/16	A
Queensland University of Technology	E46	15/16	B
The University of Auckland	E47	15/16	C
Monash University	E66	17/18	B
Monash University	A65	17/18	C
Flinders University	33	19/20	A
University of Western Australia	E20	19/20	B
University of Tasmania	E44	19/20	C
Curtin University	E14	21/22	A
National Taiwan UST	E105	21/22	B
NIT Calicut	116	23/24	A
University of Technology Sydney	E59	23/24	B
University of South Australia	E16	23/24	C
RMIT University	E88	25/26	A
University of New South Wales	A63	25/26	C

GARAGE LAYOUT

Prefix “E” in the car number designates Electric Vehicle (EV) class entries, and
Prefix “A” designates Autonomous Vehicle (AV)



ORGANISING COMMITTEE

Event Manager	Adrian Feeney
General Manager	Angela Krepcik
Facilities & Logistics Manager	Michael Lambert
Event Facilitator	TBC
Event Administration	Rose De Amicis
Manager, Rules Committee, Protest Committee	David Ford
Chief Scrutineering, Protest Committee	Michael Franks
EV Specialist, Rules Committee, Protest Committee	Grahame Holmes
Clerk of Course, Protest Committee	Nathan Tarlinton
Event Steward	Jim Stewart
EV Specialist, Rules Committee, Protest Committee	Lloyd Morrissey
Design Event Captain	Luke Phersson
Scoring Event Captain	Jason Treloar
Cost Event Captain	Jason Treloar
TI Coordinator, Faculty Advisor Representative	Rory Gover
Faculty Advisor Representative	Paul Briozzo
Presentation Event Captain	Vincent Chu
Volunteer Coordinator	Angela Krepcik
AV Demonstration Captain	Nick Owen

TRAVEL & ACCOMMODATION

LOCAL COMMUNITY ENGAGEMENT AND INFO GUIDE:

TRAVEL:

Calder Park Raceway is located approximately 30 Km from the centre of Melbourne. It is accessible from the Calder Freeway by car, with **NO WALKING ACCESS.**

CAMPING:

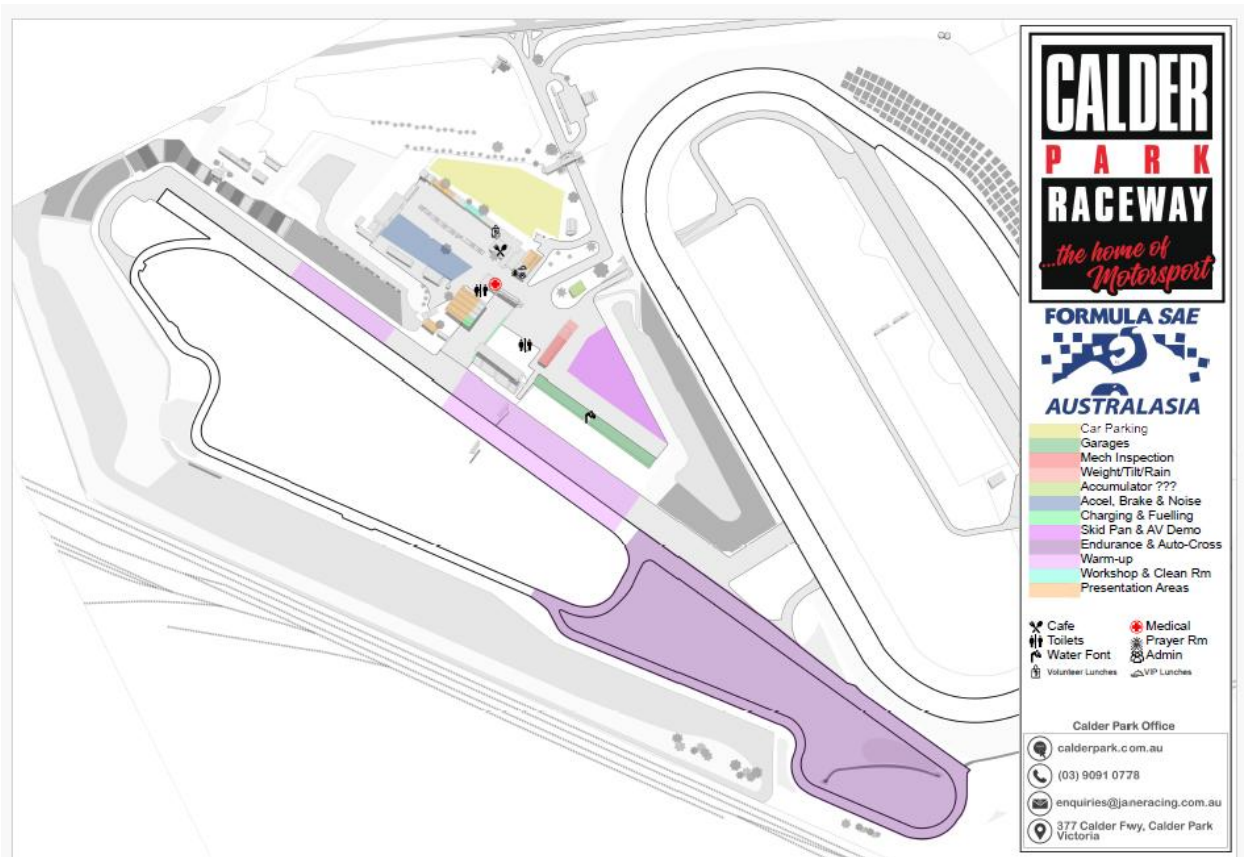
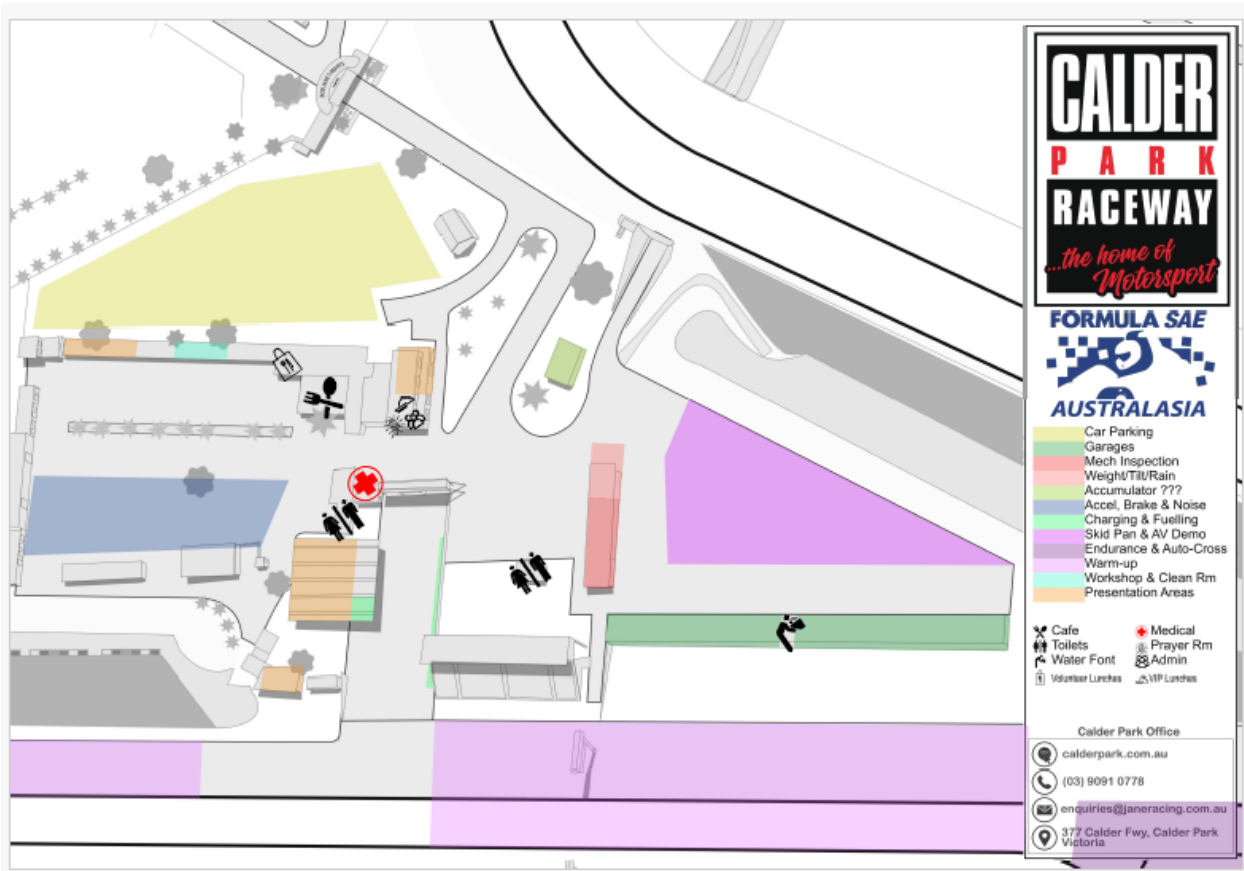
- Sundowner Caravan Park, 2057 Western Highway Rockbank. (03) 9747 1340
- Airport Tourist Village, 37 Ardlie Street, Attwood 3049. 1800 026 370
- Discovery Parks - Melbourne, 129 Ashley Street, Braybrook 3019. Ph (03) 9318 6866

CAMPING EQUIPMENT:

Camping equipment should be hired or purchased by teams. If teams plan ahead, it is often less expensive to purchase equipment through discount online retailers than to hire it. Equipment can be shipped to a host university in Australia, or directly to Calder Park Raceway. Please notify the organisers (formulasae@sae-a.com.au) if you are shipping equipment directly to the venue.

- Camping Warehouse (www.camping-warehouse.com.au/index.html)
- Kmart Australia (www.kmart.com.au)
- Aussie Disposals (www.aussiedisposals.com.au)
- Tentworld (www.tentworld.com.au/)
- BCF Australia (www.bcf.com.au/)
- Wallaroo Camping Hire (www.wallaroocampinghire.com/)
- The Equipment Library (www.equipmentlibrary.com.au/)

SITE LAYOUT



ONSITE WORKSHOP FACILITIES

Dates/Time: Refer to page 19

Location: Refer to Map on page 11

SAE-A will not be offering any workshop or welding capabilities for this year's event and as such teams are required to bring their own equipment or seek off-site facilities for such work.

Any welding may only be done in the workshop and must not be done in the pits.

A designated pit area will be provided for teams, equipped with workbenches only. Teams are expected to utilize this space responsibly and safely, mimicking the practices they would employ at their respective universities or other facilities.

Power Tools Restrictions:

No power tools are permitted within the team pits. Power tools, defined as any tool producing sparks, noise, or swarf, are restricted to the designated workshop area. This ensures a controlled environment for potentially hazardous activities.

Personal Protective Equipment (PPE) Requirements:

All individuals entering the workshop must bring suitable PPE, including:

- Fully enclosed shoes
- Long pants and shirt (preferably cotton for grinding/welding activities)
- Safety glasses
- Hearing protection
- Gloves where necessary

Safe Working Procedure:

Teams and Faculty Advisors are required to adhere to the Safe Working Procedure provided on the following page. This document outlines essential guidelines and practices to ensure a secure and controlled working environment within the workshop.

Note: Non-compliance with these guidelines may result in the team's exclusion from workshop facilities. The safety and well-being of all participants are paramount, and your cooperation is appreciated.

SAFE WORK PROCEDURE

(FORMULA SAE-A) – Safe Work Procedure WORKSHOP

DO NOT use this facility unless you have been instructed in its safe use and operation and have been given permission by your Faculty Adviser

PERSONAL PROTECTIVE EQUIPMENT



Safety glasses must be always worn in work areas.



Long and loose hair must be contained.



Do not enter this facility unless authorised.



Sturdy footwear must be worn at all times in work areas.



Close fitting/protective clothing must be worn.



Rings and jewellery must not be worn.

PRE-OPERATIONAL SAFETY CHECKS

- ✓ Locate and ensure you are familiar with all equipment operations and controls.
- ✓ Ensure the area is clean and clear of grease, oil, water, etc.
- ✓ Check workspaces and walkways to ensure no slip/trip hazards are present.
- ✓ Check that all your equipment is in good condition.

OPERATIONAL SAFETY CHECKS

- ✓ Only use properly tagged and tested electrical equipment.
- ✓ Only use equipment you are familiar with, preferably supplied by your University.
- ✓ Ensure the area is clear of people and equipment before using the workshop.

ENDING OPERATIONS AND CLEANING UP

- ✓ Leave the equipment and work area in a safe, clean and tidy state.

POTENTIAL HAZARDS AND INJURIES

- ⓘ Falling objects.
- ⓘ Trapping hazards.
- ⓘ Crushing hazards.
- ⓘ Lacerations
- ⓘ Eye damage
- ⓘ Hair/clothing getting caught in moving machine parts.

DON'T

- ✗ Do not use faulty equipment. Immediately report suspect equipment.
- ✗ Never leave the equipment running unattended.

This SWP does not necessarily cover all possible hazards associated with this facility and should be used in conjunction with other references as used at your University workshops. It is designed as a guide to be used to complement training and as a reminder to users prior to facility use.

This information is based on material published by the South Australian Department of Education and Children's Services.

Personal Protective Equipment (PPE) Requirements:

All individuals entering the workshop must bring suitable PPE, which as a minimum must include:

- Fully enclosed shoes
- Long pants and shirt (preferably cotton for grinding/welding activities)
- Safety glasses
- Hearing protection
- Gloves where necessary

ELECTRIC VEHICLE SAFETY – OVERALL

At on-site registration, teams shall provide a statement signed either by their Faculty Advisor, or an alternative appropriate University representative, that confirms their adherence to the requirements of Sections 1, 2 and 3 below, and that their vehicle, equipment and operating procedures used, meet or exceed accepted professional standards for EV safety. EV entries will not be allowed to proceed to Technical Inspection until this document has been received by an SAE-A event official.

Section 1. The team must demonstrate that they have identified, developed, implemented and upheld safety standards and a safe work culture in accordance with their applicable University policy and relevant external standards. This includes, but is not limited to:

- Undertaking electrical hazard assessments, defining electrical safe work methods and developing standard operating procedures to minimise these hazards;
- Identifying and training nominated persons to work with the electrical traction system (TS);
- Ensuring all team members who interact with the vehicle TS have first response training provided by appropriately qualified persons, that covers operational risks associated with Low Voltage AC and DC systems;
- Compliance with any legal and/or regulatory requirements applicable to both the competition site and the team's home jurisdiction;
- **Teams shall provide a detailed risk assessment for their accumulator and accumulator management processes either prior to the event or at on-site registration.**

Section 2. The Accumulator Management System (AMS) has been designed, implemented and validated to comply with all relevant safety requirements of the event rules, including in particular:

- All AMS sensors and sensor circuitry used to determine a fault condition have been designed and implemented with inherent first-degree redundancy.
- All AMS sensors and sensor circuitry used to determine a fault condition have been tested and confirmed for functionality and have been verified to trigger an appropriate response to an accumulator fault or malfunction.
- The team has verified by testing, an appropriate fault response to all conditions identified in the team's FMEA that lead to vehicle performance degradation and/or HV system shutdown.

Section 3. The team and their sponsoring university acknowledge that a failure of the vehicle TS is their responsibility and confirm that they have taken reasonable actions to independently assess their TS for safety, compliance to the intent and detail of the rules and the requirements of the competition. In particular:

- The team understands that accumulator internal temperatures will be independently passively monitored during the event, and that they will receive zero scores for all dynamic events if this monitoring identifies that they have exceeded their prescribed accumulator temperature limits during the event.

An appropriate format for this sign-off document will be provided to your Faculty Advisor.

ELECTRIC VEHICLE SAFETY- ONSITE

EMERGENCY RESPONSE: Teams shall develop an emergency response plan which addresses any potential safety issues with EV's or their accumulators, which must be available for inspection by officials during the event. The event organisers reserve the right to take any action to stop or control a fire. This may include pouring water on the accumulator and/or the team vehicle.

The entrances of pit garages shall always remain unobstructed to facilitate immediate evacuation. In the event of any incident, all team members, spectators and officials shall immediately evacuate the pit garage and surrounding area. The team's Electric Safety Responsible (ESR) person shall then notify the FSAE-A committee EV Specialist as soon as possible and follow his/her instructions. Any fire, accident, incident or near miss shall be reported to the Administration Office

CHARGING ACCUMULATORS: Teams are required to present to the EV Technical Inspectors (typically during EV-Accumulator scrutineering), a safe work method, job safety analysis, procedure documentation, nominated operators and relevant technical and safety equipment, for the safe manipulation of state of charge/charging/discharging of their accumulator pack.

Charging of accumulators is not permitted until the accumulator and charger has passed EV-Accumulator scrutineering. Accumulators may only be charged in the designated charging areas, with the accumulator container removed from the vehicle. The ESR person(s) must oversee transportation of the accumulator container between the pit garage and the charging area. The accumulator container must be labelled with the following information:

* Team name and car number * Name of ESR person and mobile phone number.

No more than three competent team members are allowed to be in a charging area while using a charging station, and at least one of these members must be a designated ESR person. Any team member supervising the charging must have the necessary knowledge/skill to act accordingly if a problem occurs. All chargers must be Class 1 devices which have been approved/marked during EV Technical Inspection. Teams must use a AS3123 compatible 32 amp, 5 pin connector to connect their charger to the installation supply. For single phase chargers, teams will be advised before the competition, which phase within their connector they should use to connect their charger.

There are six charging stations available, as shown on the site map. Each charging station has four (4) 3-phase outlets. Outlet #1 is uncoloured and may be used by any team. Outlets #2, #3, #4 are colour coded. Teams will be allocated a colour (which corresponds to the phase that they are required to use for their charger) for their accumulator charger prior to the competition, and their charger cable will be marked with this colour during their accumulator/charger inspection. If the uncoloured outlet on a charging station is in use, teams must only use an outlet with a matching colour to charge their accumulator. Otherwise, they must find another charging station that has an uncoloured or appropriately coloured outlet available.

Teams must provide their own shade coverage when they use a charging station. All shading structures must be secured while in use. Teams are responsible to properly secure their shading structure and take it down in adverse weather conditions. Teams should also place a security barrier around their charging area while it is in use to prevent unauthorised persons from entering the area.

WORKING ON ACCUMULATORS: Accumulators may only be inspected inside the pit garage. More substantial work must be carried-out in the designated EV Clean Room under the supervision of at least one ESR team member. Teams must get permission from the committee EV Specialist or his/her delegate before commencing work in the EV Clean Room. An ESR person must oversee transportation of the accumulator container between the pit garage and the EV Clean Room, and vice-versa. The entrance to the EV Clean Room must always remain

unobstructed to facilitate immediate evacuation.

ONSITE ACCUMULATOR CLEAN ROOM

Dates/Times: Refer to page 19

Location: Refer to Map on page 11

The on-site Accumulator Clean Room is located next to the on-site workshop, equipped with four (4) workbenches only. Teams are expected to utilize this space responsibly and safely, mimicking the practices they would employ at their respective universities or other facilities. A maximum of four teams only may work in the Accumulator Clean Room at any time.

Teams must get permission from the committee EV Specialists or their delegate before commencing work in the Accumulator Clean Room. Any team found working in the clean room without permission will be ordered to cease work immediately, will be in breach of the competition rules, and the Organising Committee may apply an appropriate penalty to the entire team at their absolute discretion.

Tool Restrictions:

Teams shall provide all tools used in the Accumulator Clean Room. All tools shall be appropriate to the task, and all electrical tools shall be insulated in accordance with the competition rules.

No power tools, defined as any tool producing sparks, noise, or swarf, are permitted within the Accumulator Clean Room. Electrically powered tools such as a soldering iron, or heat gun may be used.

Personal Protective Equipment (PPE) Requirements:

All individuals entering the accumulator clean room must bring suitable PPE, including:

- Fully enclosed shoes
- Long pants and shirt (preferably long sleeve)
- Safety glasses
- Electrically insulated gloves

Safe Working Procedure:

Teams and faculty advisors are required to adhere to the Safe Working Procedure provided on the following page. This document outlines essential guidelines and practices to ensure a secure and controlled working environment within the Accumulator Clean Room.

Note: Non-compliance with these guidelines may result in the team's exclusion from Accumulator Clean Room facilities. The safety and well-being of all participants are paramount, and your cooperation is appreciated.

(FORMULA SAE-A) – Safe Work Procedure

ACCUMULATOR CLEAN ROOM

DO NOT use this facility unless you have been instructed in its safe use and operation, and have been given permission by your Faculty Adviser and by a FSAE Committee EV Specialist

PERSONAL PROTECTIVE EQUIPMENT



Safety glasses must be always worn in work areas.



Long and loose hair must be contained.



Do not enter this facility unless authorised.



Sturdy, closed toe, footwear must be worn at all times in work areas.



Close fitting/protective clothing must be worn.



Rings and jewellery must not be worn.

PRE-OPERATIONAL SAFETY CHECKS

- ✓ Confirm you have approval from a FSAE Committee EV Specialist before commencing activities.
- ✓ Locate and ensure you are familiar with all equipment operations and controls.
- ✓ Ensure the area is clean and clear of grease, oil, water.
- ✓ Check workspaces and walkways to ensure no slip/trip hazards are present.
- ✓ Ensure your allocated work area is clear of other people and equipment before commencing activities.
- ✓ Check that all your equipment is in good condition.
- ✓ No work is to be undertaken unless at least 2 people are in the clean room.
- ✓ No more than 3 people to work on any one accumulator.
- ✓ All people working on an accumulator must be approved to work with electrical systems.

OPERATIONAL SAFETY CHECKS

- ✓ Only use properly tagged and tested electrical equipment.
- ✓ Only use equipment you are familiar with, preferably supplied by your University.
- ✓ Use appropriate tools for each task. Insulated tools must be used for all electrical work.
- ✓ Only work that is directly associated with your electrical accumulator is to be undertaken in this room.
- ✓ Remove the accumulator internal interconnecting links before any work is commenced, and only replace them as the last step after the work is completed.
- ✓ Electrical Safety gloves must be worn when commencing work, until the internal interconnecting links are removed to isolate the battery segments.
- ✓ Cover sections of an exposed accumulator that are not being worked on with an appropriate insulating mat.

ENDING OPERATIONS AND CLEANING UP

- ✓ Leave the equipment and work area in a safe, clean and tidy state.
- ✓ Advise the FSAE Committee EV Specialist that you have finished your accumulator work activities.

POTENTIAL HAZARDS AND INJURIES

- ⓘ Electrical Fire
- ⓘ Electrocutation
- ⓘ Flash Burns
- ⓘ Eye damage
- ⓘ Falling objects.
- ⓘ Trapping hazards.
- ⓘ Crushing hazards.
- ⓘ Lacerations
- ⓘ Hair/clothing getting caught in moving machine parts.

DON'T

- ✗ Do not use faulty equipment. Immediately report suspect equipment.
- ✗ Never leave the equipment running unattended.

COMPLIANCE WITH NFPA70E

All work on accumulators shall be conducted in accordance with NFPA70E safe working practices relating to electrocution and arc flash hazards.

Appropriate Personal Protective Equipment (PPE) must be worn when working on accumulator, of an appropriate Class to suit the identified arc flash risk level for work being conducted.

RESPONSE TO THERMAL INCIDENT

In the event of an accumulator thermal incident, all personnel SHALL immediately evacuate the affected area and declare an emergency. No attempt should be made to extinguish an accumulator in the first instance.

This SWP does not necessarily cover all possible hazards associated with this facility and should be used in conjunction with other references as used at your University workshops. It is designed as a guide to be used to compliment training and as a reminder to users prior to facility use.

This information is based on material published by the South Australian Department of Education and Children's Services.

(FORMULA SAE-A) – Safe Work Procedure

ACCUMULATOR INSPECTION ROOM

DO NOT use this facility unless you have been instructed in its safe use and operation, and have been given permission by your Faculty Adviser and by a FSAE Committee EV Specialist

PERSONAL PROTECTIVE EQUIPMENT



Safety glasses must be always worn in work areas.



Long and loose hair must be contained.



Do not enter this facility unless authorised.



Sturdy, closed toe, footwear must be worn at all times in work areas.



Close fitting/protective clothing must be worn.



Rings and jewellery must not be worn.

PRE-OPERATIONAL SAFETY CHECKS

- ✓ Confirm you have approval from a FSAE Committee EV Specialist before commencing activities.
- ✓ Locate and ensure you are familiar with all equipment operations and controls.
- ✓ Ensure the area is clean and clear of grease, oil, water.
- ✓ Check workspaces and walkways to ensure no slip/trip hazards are present.
- ✓ Ensure your allocated work area is clear of other people and equipment before commencing activities.
- ✓ Check that all your equipment is in good condition.
- ✓ No work is to be undertaken unless at least 2 people are in the clean room.
- ✓ No more than 3 people to work on any one accumulator.
- ✓ All people working on an accumulator must be approved to work with electrical systems.

OPERATIONAL SAFETY CHECKS

- ✓ Only use properly tagged and tested electrical equipment.
- ✓ Only use equipment you are familiar with, preferably supplied by your University.
- ✓ Use appropriate tools for each task. Insulated tools must be used for all electrical work.
- ✓ Only work that is directly associated with your electrical accumulator is to be undertaken in this room.
- ✓ Electrical Safety gloves must be worn when commencing work, until the internal interconnecting links are removed to isolate the battery segments.
- ✓ Cover sections of an exposed accumulator that are not being worked on with an appropriate insulating mat.

ENDING OPERATIONS AND CLEANING UP

- ✓ Leave the equipment and work area in a safe, clean and tidy state.

POTENTIAL HAZARDS AND INJURIES

- ⓘ Electrical Fire
- ⓘ Electrocutation
- ⓘ Flash Burns
- ⓘ Eye damage
- ⓘ Falling objects.
- ⓘ Trapping hazards.
- ⓘ Crushing hazards.
- ⓘ Lacerations
- ⓘ Hair/clothing getting caught in moving machine parts.

DON'T

- ✗ Do not use faulty equipment. Immediately report suspect equipment.
- ✗ Never leave the equipment running unattended.

COMPLIANCE WITH NFPA70E

All work on accumulators shall be conducted in accordance with NFPA70E safe working practices relating to electrocution and arc flash hazards.

Appropriate Personal Protective Equipment (PPE) must be worn when working on accumulator, of an appropriate Class to suit the identified arc flash risk level for work being conducted.

RESPONSE TO THERMAL INCIDENT

In the event of an accumulator thermal incident, all personnel SHALL immediately evacuate the affected area and declare an emergency. No attempt should be made to extinguish an accumulator in the first instance.

This SWP does not necessarily cover all possible hazards associated with this facility and should be used in conjunction with other references as used at your University workshops. It is designed as a guide to be used to compliment training and as a reminder to users prior to facility use.

This information is based on material published by the South Australian Department of Education and Children's Services.

FORMULA SAE-A STUDENT (FS) TEAMS

Each global competition will have their own administrative and event control requirements, which are independent of vehicle design and construction. The FSAE-A administrative requirements and event procedures are per the Formula SAE International Rules, except where modified or complemented by the FSAE-A Addendum. Regardless of the rules used for the design/build of vehicles, all teams must comply with these requirements.

FSAE-A is intended as a university supported and educational event for student teams. University oversight of a team's preparation and entry is considered essential, and therefore the Faculty Advisor (FA) is a key role. As the Formula Student (FS) Rules no longer refer to a FA, the following will apply to vehicles using just the FSG Rules for the FSAE-A competition.

If the university has had a FA in place, then he/she should attend the FSAE-A event, and the other FA notes included in the International Rules and FSAE-A Addendum will apply. If the university has not had a FA in place, then for the team entry to be accepted, a Statutory Declaration (a legally binding form or authority), as described below, signed by a permanently employed member of the University's Engineering School or Faculty, must be provided with the team's registration. A university employee must also accompany the team to fulfil the role of FA at the event. The declaration must state that the design/build of the vehicle has been wholly undertaken by team members, without outside assistance from university staff, external professionals or sponsors. The person signing must be at the minimum level of Associate Professor.

GROUP A VEHICLES: Vehicles constructed to comply with both the 2025 FS and 2025 FSAE International Rules. These vehicles should comply with most design/build requirements. Only the additional or modified requirements as detailed in the FSAE-A Addendum need to be addressed by teams to ensure compliance.

GROUP B VEHICLES: Vehicles constructed only taking account of the 2025 FS Rules. There is no published list of items that identify the exact variations which may affect the ability of vehicles built to the FS rules to comply with the FSAE International rules. It is therefore up to individual teams to assess their compliance.

It is recommended that teams develop a summary list of all variances they can identify and submit this listing, with as much detail as possible, to the FSAE-A Rules Committee for assessment of acceptability. Non-compliance in some areas will not mean automatic disqualification. Items affecting vehicle safety will be of most concern. For example, it is known the FS Rules inadvertently omitted the minimum width requirement at the base of the main roll hoop, which affects rollover safety.

GROUP A & B VEHICLES: For any specific rules related question or interpretation, teams should contact the event organisers formulasae@sae-a.com.au or submit a formal rules enquiry as outlined in the Addendum. If SES submissions have been previously approved at other competitions, then this should facilitate approval at the FSAE-A event.

Submission of as much detail as possible will assist the review and acceptance by the FSAE-A Rules Committee.

EVENT SCHEDULE

Wednesday 10 Dec 2025	
15:00 – 17:00	Team Registration & Bump In
17:00	Team Welcome
18:00 – 18:30	Cost Event Briefing
18:00	Gates Closed

Thursday 11 Dec 2025	
7:00	Gates Open
7:30	Volunteer Sign In & Breakfast
8:00	Team Registration, Technical Inspection, Cost Event Open
12:00	Volunteer Lunch All Events Closed
13:00	Technical Inspection, Cost Event, Fuel (on request), Brake & Noise Open
17:00	Technical Inspection, Cost Event, Fuel, Brake & Noise Closed
19:00	Gates Closed

Friday 12 Dec 2025	
7:00	Gates Open
7:30	Volunteer Sign In & Breakfast
8:00	Business Presentation, Design, Technical Inspection, Practice Track, Brake & Noise, Fuel (on request) Open
12:00	All Events Closed VIP Tour & Lunch Volunteer Lunch
13:00	All Event Open
17:00	All Events Closed
18:00	Design finalists announced Careers Expo
19:30	Careers Expo Closed
20:00	Gates Closed

Saturday 13 Dec 2025	
7:00	Gates Open
7:30	Volunteer Sign In & Breakfast
8:00	Volunteer Briefing
8:30	Driver Briefing Technical Inspection, Fuel, Practice Track, Brake & Noise Open
9:00	Skid Pad Event Open
11:30	Skid Pad, All Events Closed Faculty Advisor Meeting
	Volunteer Lunch

Saturday 13 Dec 2025 (continued)	
12:30	Technical Inspection, Fuel, Practice Track Brake & Noise, Acceleration Event Open
13:00	Cost & Presentation Results Released
15:00	Acceleration Event Closed
15:30 – 17:00	Autonomous Vehicle Track Demonstration Event
16:00	Practice Track Closed
17:00	Static Events Feedback Session Driver Briefing & Mandatory Course Walk All Events Closed
18:00 – 19:00	Design Event Finals
19:00	Gates Closed

Sunday 14 Dec 2025 (continued)	
7:00	Gates Open
7:30	Volunteer Sign In & Breakfast
8:00	Volunteer Briefing
8:30	Driver Briefing Technical Inspection Open
9:00	Autocross Event Open Practice, Fuel Open AV Design Lite
11:30	Autocross Closed Practice Track, Fuel, Technical Inspection Closed Volunteer Lunch
12:30	Practice Track, Fuel (on request), Technical Inspection Open
13:00	Endurance Event Open Parc Ferme Open
16:30	Endurance Event Closed Practice Track Closed
17:00	Fuel, Parc Ferme Closed
17:30	Team Photos
18:15	Awards Ceremony
19:30	Gates Closed

STATIC EVENTS & TECHNICAL INSPECTION SCHEDULE

Car Number	UNIVERSITY	GARAGE	ACTIVITY							
			EV-STATIC	MECHANICAL	ACCUM	EV-FUNCT	AV INSPECT	COST	DESIGN	PRESENTATION
7	Edith Cowan University	9/10 A		Thurs 8:00				Thurs 13:00	Fri 10:00	Fri 14:00
21	Tokyo Denki University	7/8 A		Thurs 16:00				Thurs 8:00	Fri 14:00	Fri 10:00
33	Flinders University	19/20 A		Thurs 16:00				Thurs 9:00	Fri 15:00	Fri 9:00
61	Australian National University	13/14 A		Thurs 8:00				Thurs 11:00	Fri 13:00	Fri 16:00
70	Alfaisal University	15/16 A		Thurs 8:00				Thurs 14:00	Fri 8:00	Fri 15:00
111	Griffith University	13/14 B		Thurs 8:00				Thurs 15:00	Fri 9:00	Fri 11:00
112	Institut Teknologi Bandung	17/18 A		Thurs 10:00				Thurs 16:00	Fri 11:00	Fri 13:00
116	NIT Calicut	23/24 A		Thurs 16:00				Thurs 10:00	Fri 16:00	Fri 8:00
E105	National Taiwan UST	21/22 B	Thurs 10:00	Thurs 11:00	Thurs 8:00	Fri 11:00		Thurs 14:00	Fri 8:00	Fri 16:00
E13	University of Canterbury	9/10 C	Thurs 13:00	Thurs 14:00	Thurs 8:00	Fri 14:00		Thurs 10:00	Fri 16:00	Fri 10:00
E14	Curtin University	21/22 A	Thurs 10:00	Thurs 11:00	Thurs 8:00	Fri 11:00		Thurs 14:00	Fri 8:00	Fri 13:00
E15	UNSW ADFA	7/8 C	Thurs 8:00	Thurs 9:00	Thurs 14:00	Fri 10:00		Thurs 11:00	Fri 13:00	Fri 8:00
E16	University of South Australia	23/24 C	Thurs 11:00	Thurs 13:00	Thurs 9:00	Fri 13:00		Thurs 15:00	Fri 15:00	Fri 9:00
E17	Swinburne University of Technology	5/6 B	Thurs 11:00	Thurs 13:00	Thurs 9:00	Fri 13:00		Thurs 15:00	Fri 9:00	Fri 11:00
E20	University of Western Australia	19/20 B	Thurs 11:00	Thurs 13:00	Thurs 9:00	Fri 13:00		Thurs 15:00	Fri 15:00	Fri 9:00
E22	The University of Sydney	7/8 B	Thurs 13:00	Thurs 14:00	Thurs 8:00	Fri 14:00		Thurs 10:00	Fri 16:00	Fri 10:00
E25	The University of Waikato	21/22 C	Thurs 10:00	Thurs 11:00	Thurs 14:00	Fri 8:00		Thurs 8:00	Fri 11:00	Fri 14:00
E3	University of Newcastle	11/12 A	Thurs 8:00	Thurs 9:00	Thurs 13:00	Fri 9:00		Thurs 11:00	Fri 13:00	Fri 15:00
E42	The University of Queensland	11/12 B	Thurs 13:00	Thurs 14:00	Thurs 10:00	Fri 14:00		Thurs 16:00	Fri 16:00	Fri 8:00
E44	University of Tasmania	19/20 C	Thurs 11:00	Thurs 13:00	Thurs 9:00	Fri 11:00		Thurs 16:00	Fri 9:00	Fri 13:00
E46	Queensland University of Technology	15/16 B	Thurs 9:00	Thurs 10:00	Thurs 15:00	Fri 11:00		Thurs 13:00	Fri 14:00	Fri 9:00
E47	The University of Auckland	15/16 C	Thurs 14:00	Thurs 15:00	Thurs 11:00	Fri 8:00		Thurs 9:00	Fri 10:00	Fri 13:00
E59	University of Technology Sydney	23/24 B	Thurs 10:00	Thurs 11:00	Thurs 14:00	Fri 9:00		Thurs 8:00	Fri 11:00	Fri 14:00
E66	Monash University	17/18 B	Thurs 8:00	Thurs 9:00	Thurs 11:00	Fri 8:00		Thurs 14:00	Fri 10:00	Fri 15:00
E68	The University of Melbourne	5/6 C	Thurs 13:00	Thurs 14:00	Thurs 10:00	Fri 14:00		Thurs 16:00	Fri 9:00	Fri 11:00
E74	Institut Teknologi Sepuluh Nopember	9/10 B	Thurs 9:00	Thurs 10:00	Thurs 15:00	Fri 10:00		Thurs 13:00	Fri 14:00	Fri 8:00
E8	The University of Adelaide	13/14 C	Thurs 8:00	Thurs 9:00	Thurs 13:00	Fri 9:00		Thurs 11:00	Fri 11:00	Fri 14:00
E85	University of Wollongong	5/6 A	Thurs 9:00	Thurs 10:00	Thurs 15:00	Fri 10:00		Thurs 13:00	Fri 14:00	Fri 16:00
E88	RMIT University	25/26 A			Thurs 13:00			Thurs 10:00	Fri 13:00	Fri 15:00
A48	The University of Queensland	11/12 C	Thurs 14:00	Thurs 15:00	Thurs 11:00	Fri 10:00	Fri 15:00			
A63	University of New South Wales	25/26 C	Thurs 14:00	Thurs 15:00	Thurs 10:00	Fri 8:00	Fri 13:00	Thurs 8:00	Fri 10:00	Fri 16:00
A65	Monash University	17/18 C	Thurs 14:00	Thurs 15:00	Thurs 11:00	Fri 9:00	Fri 14:00			

AWARDS CEREMONY

Electric Vehicle – Class Winner	1st Place	2nd Place	3rd Place
Acceleration			
Autocross			
Business Presentation			
Cost			
Efficiency			
Engineering Design			
Skid Pad			

Internal Combustion Class	1st Place	2nd Place
IC Class Winner		
Acceleration		
Autocross		
Business Presentation		
Cost		
Efficiency		
Engineering Design		
Endurance		
Skid Pad		

Specialty Awards	Awards
MA Inspiring Motorsport Awards	Awarded to team that shows the best spirit
LEAP Award for Most Innovative Use of Simulation	Awarded to team that best uses simulation
SAE-A Harry Watson Award	Awarded to team that makes a distinct contribution
Caterpillar Automated Vehicle Drive Award	Awarded to Formula SAE participants who demonstrate excellence in automation and engineering
Technical Inspection Award	Awarded to best presented team at Technical Inspection

VEHICLE INSPECTION PANEL

Class-specific base stickers are provided for IC, EV and AV. Base panels will be located along the centreline of the nose cone, as per the instructions within the FSAE-A Rules Addendum.

The panels track the progress of the vehicles through the mandatory technical inspections. Vehicle mass and Fuel Capacity/Accumulator Voltage will be recorded at Tilt Test and Fuel Station/EV Functional Inspection, respectively, and noted on the base sticker. Coloured inspection stickers will be issued to the teams upon successfully passing the relevant inspections, with event inspection stickers applied to the base sticker by the team. Teams must have all relevant inspection stickers applied to their base sticker to be able to compete in any Dynamic Events. Dual purpose vehicles may elect to forego full AS and EBS inspections and still compete in Dynamic Events. In such cases, the team will need to complete a relevant portion of AS before being issue a sticker confirming their reversion to EV only.

COMPLETED AND FILLED AV STICKER

The AV Sticker is a rectangular panel with a blue header. The header contains a car icon with a radio wave, the text 'AV CLASS', a 'DUAL PURPOSE' checkbox, and two input fields for 'CAPACITY' (300.4) and 'VDC' (244.8). The right side of the header features the 'FORMULA SAE AUSTRALASIA' logo. Below the header is a section labeled 'SAE-A OFFICIAL USE ONLY' containing eight colored icons: Mech Inspection (blue), Rain (light blue), Tilt Test (green), Brake Test (yellow), AS Inspection (dark green), Accumulator (orange), EV Static (purple), EV Functional (red), and EBS Test (pink).

COMPLETED EV STICKER

The EV Sticker is a rectangular panel with a blue header. The header contains a car icon with a lightning bolt, the text 'EV CLASS', and two empty input fields for 'CAPACITY' and 'VDC'. The right side of the header features the 'FORMULA SAE AUSTRALASIA' logo. Below the header is a section labeled 'SAE-A OFFICIAL USE ONLY' containing seven colored icons: Mech Inspection (blue), Rain (light blue), Tilt Test (green), Brake Test (yellow), Accumulator (orange), EV Static (purple), and EV Functional (red). The EBS Test icon is missing.

IC BASE PANEL

The IC Base Panel is a rectangular panel with a blue header. The header contains a car icon, the text 'IC CLASS', and two empty input fields for 'CAPACITY' and 'VDC'. The right side of the header features the 'FORMULA SAE AUSTRALASIA' logo. Below the header is a section labeled 'SAE-A OFFICIAL USE ONLY' containing four greyed-out icons: Mech Inspection, Rain, Tilt Test, and Brake Test. Each icon has a 'NOT COMPLETE' label and 'EV / IC' text below it.

TECHNICAL INSPECTION

Technical Inspection Coordinator: Rory Gover

Dates/Times: Refer to page 20

Technical Inspection (TI) will establish if the vehicle complies with (and satisfies the intent of) the competition rules. Teams should attend their scheduled inspection time and bring with them all the necessary equipment and documentation (must be hard copy). It will include:

- Vehicle, push bar, fire extinguishers, and one set of each tyre type (dry and wet). (Mech)
- All driver safety equipment. (Mech)
- A completed (self-assessment) technical inspection form. (All inspections)
- Structural equivalency spreadsheet (SES). (Mech)
- Impact attenuator report with the tested sample, including bulkhead sample. (Mech)
- Copies of any submitted rules enquiries, along with the answer(s) you received. (All inspections)
- Approval to use Electronic Throttle Control (ETC) and ETC FMEA, if relevant. (Mech)
- Electronic Throttle Control (ETC) FMEA - IC class only, if relevant. (Mech)
- Electrical system form, Electrical systems officer form, Electrical systems advisor form. (EV Static)
- Sample of firewall to check AL thickness. (EV Static)
- Sample of accumulator container floor and top/side walls if not made from metal. (EV Static)
- Accumulator, Accumulator Cart, Charger & ancillaries, HV tools, Design docs and datasheets. (Accumulator)
- Autonomous Systems Form. (AS Inspection)
- Autonomous Wiring Diagram. (AS Inspection)
- Autonomous Hydraulic/Pneumatic Systems Documents. (AS Inspection)
- A safe method for securing the vehicle while the wheels are driven above ground (e.g. stands) as prescribed in the Technical Inspection Sheet. (EV Functional)

Teams will be provided with a bound, hard copy of Technical Inspection documentation upon registration. Teams can obtain a copy of this documentation online to complete their vehicle self-assessment (<https://www.saea.com.au/rules-documents-templates>). Teams should refer to this documentation for the complete checklist of equipment/documentation required for Technical Inspection for all classes.

Teams will be allocated a specific time to present their car for Technical Inspection. This time must be adhered to, and no team is to take other teams allocated time unless approved by the Technical Inspection Coordinator. The timings of scheduled inspection will be marshalled. Teams are to arrive at Technical Inspection no earlier than 10 minutes before their scheduled inspections.

No videography, photography or audio recording by teams or public is permitted within Technical Inspection events. The Technical Inspection Coordinator reserves the right to request teams to cease intrusive recording in the vicinity of Technical Inspection. Technical Inspection volunteers may keep confidential record (including photography) of vehicles.

No work is to be conducted on vehicles in the vicinity of the Technical Inspection building or in nearby thoroughfares. Re-scrutineering bays for team undertaking minor works may be made available with approval from the relevant Inspection Captain and Technical Inspection Coordinator.

Petitions or appeals from teams regarding Technical Inspection assessments must be made to the responsible Event Captain in the first instance. If further clarification is required, the Chief Scrutineer may be consulted. If a team seeks to appeal further, they may lodge a protest to the Protest Committee.

TECHNICAL INSPECTION – CONT

Prior to departing from home, each team must complete their own Vehicle Inspection check, using the appropriate TII documentation. This documentation must be presented at registration and first inspection.

Tech Inspection Stages:

EV – Static (Scheduled Thursday)

1. Electrical Technical Inspection (PART 1) Static Inspection

Accumulator Inspection (Scheduled Thursday)

1. Electrical PPE inspection
2. Accumulator & charger technical inspection
3. Thermal Strip Installation

Mechanical Inspection (Scheduled Thursday)

1. Static Mechanical Technical Inspection
2. Tallest Driver only Egress (IC and EV Only)

Tilt Table (Unscheduled, Thursday/Friday)

1. Fuelling (IC Only, Friday Only)
2. Weighing
3. Tilt Table Test

EV – Functional (Scheduled Friday)

1. Electrical Technical Inspection (PART
2. Functional Inspection and Demonstration Verification of Energy Meter.

AS Inspection (Scheduled Friday)

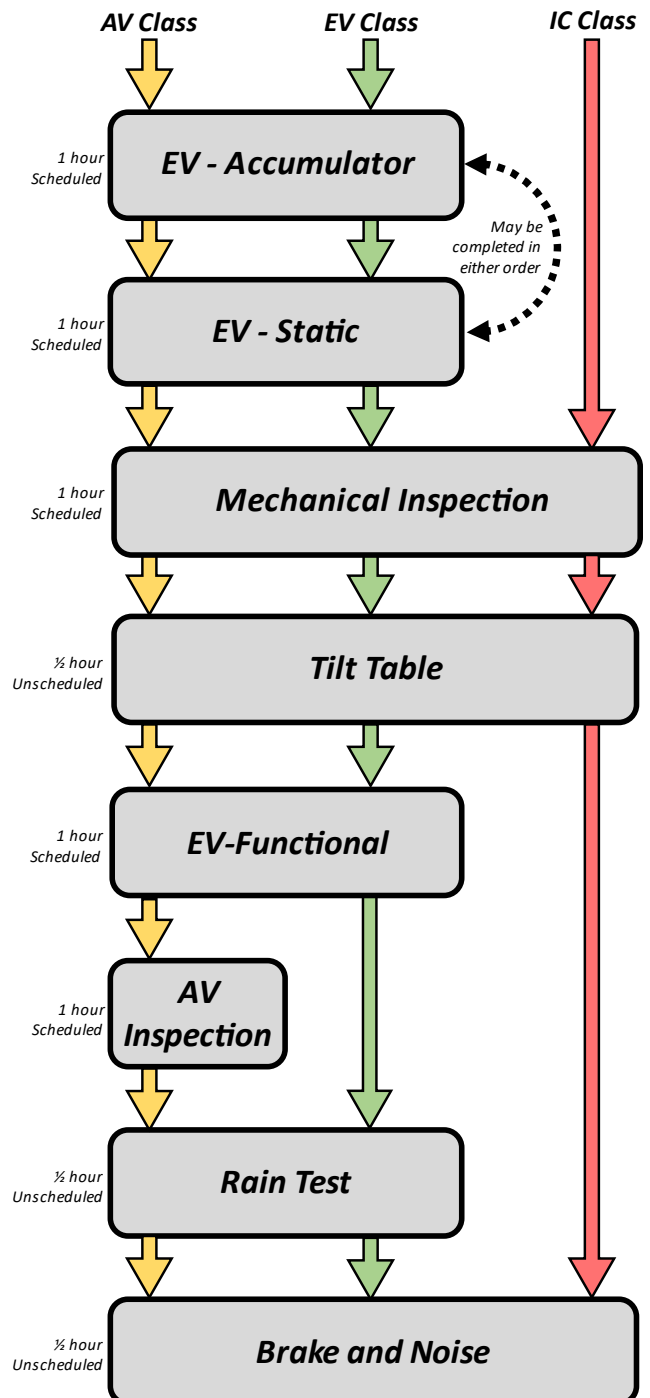
1. Autonomous Systems Technical Inspection

EV – Rain Test (Unscheduled, Friday)

1. Rain Test

Brake and Noise Area (Unscheduled, Friday)

1. Noise Test (IC Only)
2. Brake Test (IC and EV Only)
3. EBS Test (AV Only)



Note: Driver package and helmet clearance may be rechecked for any driver in the staging area at the start of Dynamic events.

WEIGH STATION & TILT TABLE

Event Captain: Michael Franks

Dates/Times: Refer to page 19

WEIGH STATION: All teams shall have their vehicles weighed, preferably after passing Mechanical Inspection. Vehicles are to be complete and ready to run, i.e., with coolant, lubricants and brake fluid, but without fuel. EV's must have the accumulator installed. Results will be recorded on the vehicle's inspection sticker panel.

In the event that a vehicle has not passed Technical Inspection prior to the team's designated time for Design Event judging, the vehicle must still pass through the Weigh Station (without fuel) before commencing the Design event. All teams are responsible for allowing enough time for this activity. Vehicles may be re-weighed during Parc Ferme at the discretion of the organisers. Major discrepancies will be brought to the attention of the Protest Committee.

TILT TABLE: The Tilt Table ensures vehicles can maintain a lateral acceleration of 1.7g without rolling over. A fluid integrity test is also performed to ensure no leaks are present. IC Teams must first proceed to the Fuel Station and then return to the Tilt Table. The table will initially be tilted to an angle of 45 degrees with the tallest driver in the vehicle. If the upper wheels remain on the table, then the angle will be increased to 60 degrees. The upper wheels must remain in contact with the table to pass the test. Vehicles will be tested in both directions. Note: For presentation at Tilt Table, all Vehicles must be filled with all fuel, coolant and any other fluids at their 'Full' level.

FUEL STATION

Event Captain: Dan Marshall

Dates/Times: Refer to page 19

The Fuel Station will provide premium gasoline (RON 98) and E85 fuel. Teams should clearly designate the required fuel near the fuel filler neck. The first time a vehicle is fuelled, the total capacity of the fuel system will be recorded on the inspection sticker panel. Vehicles will not be fuelled until passing Mechanical Inspection. The vehicle must proceed immediately to the Tilt Table to perform a fluid integrity test. The total capacity of the fuel system may be re-checked during Parc Ferme and/or subsequent refuelling. Major discrepancies will be brought to the attention of the Clerk of Course and Protest Committee.

BRAKE AND NOISE TEST

Event Captain: Jon Arbour

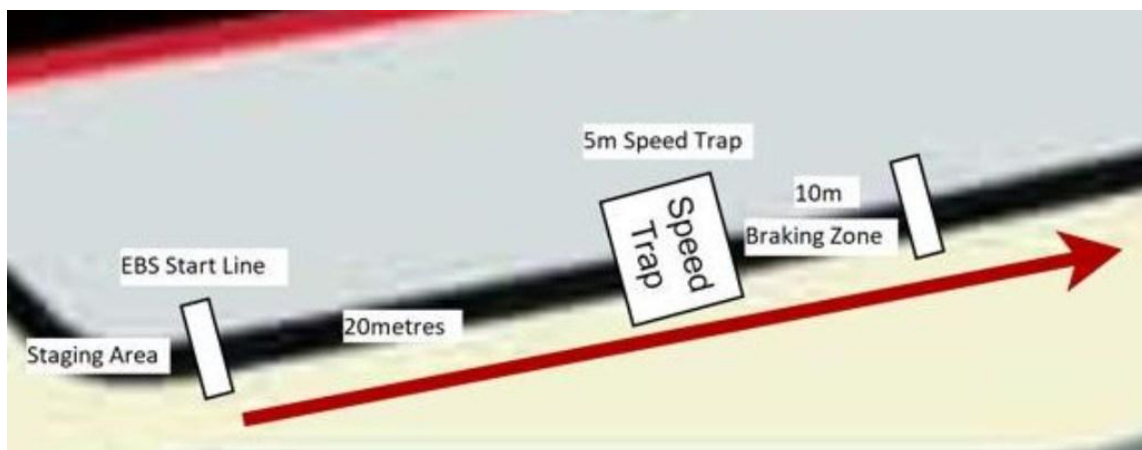
Dates/Times: Refer to page 19

The Brake and Noise tests assess that vehicles have an operational braking system, and that they do not exceed the maximum allowed noise level. Vehicles are not permitted to perform these tests until they have passed Technical Inspection and Tilt Table. Brake and Noise tests may be done in any order.

NOISE TEST PROCEDURE (IC Only): The static sound level test shall occur on a warmed-up vehicle. The measurement must not exceed 103 dBC with the engine idling, or 110 dBC at the designated test speed. The noise level can be re-measured at any time during the event. Penalties may be assessed if the sound level exceeds the mandated maximum any time after passing the test.

BRAKE TEST PROCEDURE: The driver must accelerate until reaching the braking area. Once inside this area, the driver must apply the brakes and demonstrate that all four wheels can lock-up simultaneously. The brake light intensity will also be assessed. In damp conditions, the brake test will be conducted at the discretion of the Event Captain. Vehicles may be asked to return to the brake test for recertification at any stage during the event. Note: For the Brake Test, the tyre pressures must be set within the range defined in Clause IN.12.4 of the Local Rules Addendum.

AV EMERGENCY BRAKE TEST PROCEDURE (EBS): Start-up Procedure for Autonomous Running, no additional equipment (e.g. laptop, jack-up device, pressure tank, etc.) is allowed to start up the vehicle at the staging/starting line. If the vehicle does not enter “AS Ready” state within 1 min after being staged, the team may be sent back to the preparation area by the officials. The vehicle may only be staged with the steering system in straight position. The vehicle may be pushed from the preparation area to the start line with activated LVS. The ASB may be energized already in the preparation area. The ASMS may only be switched on by the ASR after approval from an official at the starting Line. During the brake test, the vehicle must accelerate in autonomous mode up to at least 40 km/h within 20m. From the point where the RES is triggered, the vehicle must come to a safe stop failure to do so will result in a failure.



RAIN TEST

Event Captain: Lloyd Morrissey

Dates/Times: Refer to page 19

The Rain Test checks the vehicle's ability to operate safely in wet weather conditions.

The vehicle will be lifted so that the wheels cannot touch the ground and powered on. Water will be sprayed on the vehicle from all directions to simulate heavy rainfall and spray. Water will be sprayed for 60 seconds, and then the vehicle will remain on for a further 60 seconds.

The vehicle must remain powered on without any misbehaviour or faults becoming present during the test.

Teams should bring their vehicle in race ready condition, a set of secure stands to support the vehicle, safety equipment and technical inspection documents. Some dry towels are also a good idea.

The rain test is unscheduled and can be attempted after the vehicle has passed EV functional.

ACCUMULATOR INSPECTION

Event Captain: Lloyd Morrissey

Dates/Times: Refer to page 19

The Accumulator Inspection is one of the major steps in the overall EV Technical Inspection process. Teams will be given a scheduled time for their accumulator inspection separate to the overall vehicle inspection.

The accumulator inspection will be in a dedicated inspection area where the accumulators will be opened and inspected in detail.

Teams should bring their accumulator, charger, technical inspection documentation, datasheets for major components, insulated tools and any other equipment or documentation required to safely open and inspect their accumulator. The vehicle itself, drivers' equipment etc. are not required.

Teams should expect to be asked by the technical inspectors to demonstrate the safety features of their accumulator, with a particular focus on prevention of electric shock, short circuits and fires. A major focus will be on quality of construction and electrical insulation.

Inspection of accumulator PPE, tools and charger will be conducted in the area immediately external to the main entrance of the building. Inspection of the accumulator may not begin until PPE and tools inspection is completed.

Due to the nature of the accumulator inspection, only essential personnel from the team will be allowed into the inspection area, and no public viewing is allowed.

The ESO must be present at all times during the accumulator inspection.

DESIGN EVENT

Event Captain: Luke Phersson

Dates/Times: Refer to page 19

The Design Event assesses the team's knowledge of Engineering Design and Project Management concepts. Teams will present their vehicles in the judging area at the time designated in the static events schedule. All teams must provide prior to the competition a Design Briefing, 3-View Drawings, and Specifications Sheet. Teams that do not submit these documents will be disqualified and receive zero points for the Design Event.

New for 2025, will be a Design Changes Summary, which will be a google forms submission to give a brief overview on the level of design change year to year. The link is available in the Design Event Overview document on the SAE-A website.

The intention of this submission is to try to capture the major changes to your vehicle's year-to-year in an effort to avoid arbitrary expensive chassis re-manufacture to avoid historical 2nd year vehicle penalties. To be clear, this document will not be used for any scoring this year, it's just to create a point of reference for future years. However, failure to submit, or submitting an incomplete entry may result in penalties this year.

Design judging will take place across several areas - please see the SAE-A website for the Design Event Overview document containing the scoring rubric for a detailed breakdown. Each team should have at least one representative who is prepared to discuss each major system individually. Each judge has a different area of expertise and will seek out the team member responsible for his/her area of the vehicle. Roving judges may also briefly assess cars across judging groups, while noting interesting vehicle features or flaws. In general, judges will be looking for evidence of:

- Structured decision-making processes.
- Sound understanding of the vehicle level design trade-offs.
- A structured breakdown of design goals from vehicle level cascading down to component level.
- Evidence of appropriate design decisions being made for the resources available.
- Evidence of a structured vehicle testing program with defined goals and validated conclusions.
- Vehicle manufacturability and serviceability.

Teams may only be allowed to have ten students inside the judging area at any one time but can freely interchange team members - this will be communicated closer to the event if it will be enforced. Any delaying of the judges' movements to the next team will result in a penalty.

New for 2025, we will be running Design Finals after the Dynamic Events on the Saturday. This will enable improved judging quality between the different Judging Streams. It will run similar to the standard Design Event, however, will in general go deeper into topics to test the depth of the teams understanding. It will be a consistent group of experienced judges for all teams in Design Finals.

COST EVENT

Event Captain: Jason Treloar

Dates/Times: Refer to page 19

The Cost and Manufacturing event, typically referred to as simply “the Cost event”, exists to expand the scope of the competition past that of a purely design focused challenge. Its aim is to provide students an introductory view into the downstream processes in a graduate automotive engineering business, particularly those past the design and prototyping phase, and the realities of cost-to-performance engineering trade-offs. The event focuses on 4 core engineering sub-disciplines, being cost engineering, process engineering, manufacturing engineering, and technical release drafting. To that end, the cost event assesses teams’ collective knowledge of manufacturing engineering and capability and project management with respect to cost-based parameters, and decision trade-offs made between the vehicle manufacturing cost, profitability for the business enterprise per rule GR.1.4.1, and the dynamic performance of the vehicle.

COST SCORING AND SUBMISSIONS BREAKDOWN:

The Cost and Manufacturing event is broken down into 3 main scored subsections, being the cost report, worth 80% of the teams’ points, the cost scenario worth 15% of the teams’ points, and the cost challenge, been worth the remaining 5% of the teams’ points, adding to a total of 100 points possible for the cost event. Teams which fail to submit a cost report by the final cut-off date of the cost report submission are excluded from the cost report scored section of the competition, and only able to achieve up to 20 points for the cost event.

Teams have several submissions and actions which are completed through the year for the cost event. In order of delivery, these are the cost report and supporting documentation submission, the cost amendment report submission, the cost briefing, then the cost and manufacturing physical event, where teams will be allowed full control over their time allocation within an allocated time slot in order to challenge accuracy issues or concerns found in the cost report, complete the presentation of the results of the cost scenario, and complete the cost challenge, demonstrating their capacity to reverse engineer the processes required to build a unique item.

Scoring for these subsections can be found in the FSAE-A Rules addendum. For the cost report element, at a high-level view, this takes the overall reported vehicle cost, the accuracy level estimation from the cost judges of the report, focusing only on a specific subset of systems, and the documentation level for a specific system or group of systems, which is then ranked to determine the final score for the team. The cost scenario and cost challenge are both determined at the discretion of the cost judge based on a set of criteria made surrounding the elements expected for the specific event subsection section.

COST EVENT (continued)

COST EVENT RUNTIME BRIEF:

Teams are allocated in 1-hour slots for the cost event, split in 4 streams, as found on page 18. These allocations are randomly generated, with no preference provided for any team. Teams must attend and present during this allocated slot, no provisions for changes in allocation or runtime changes are allowed, excluding exceptional circumstances. The first 7.5 minutes of this time is allocated for teams to set up the area for the cost event, including partial disassembly of vehicle to show obstructed components or areas otherwise not visible. Due to parallel scrutineering activities, teams competing in the Electric or Autonomous classes do not need to come with an accumulator present in their vehicle, but the rest of the vehicle should be as complete as possible. Teams then have 45 minutes to complete the cost event, presenting their cost scenario result, the cost challenge, and challenging any points of contention from their specific cost accuracy sheet. Once the team has finished their cost event activities, or if the 45 minutes concludes, teams are expected to fully vacate the area in the remaining 47.5 minutes of their allocated slot. If a team is still presenting at the 40-minute mark, they will be advised they have 5 minutes left. The event will hit a hard stop at the 45-minute mark.

COST EVENT PERSONEL AND EQUIPMENT CONSTRAINTS:

For both safety and mobility purposes, teams are not to have more than 8 team members in the judging area at a time during the cost event activities (i.e. after setup and before pack down). Teams may setup sun or rain protection structures immediately at the front of the cost judging bay, however, must ensure that movement in the general area is not affected for both their team members, adjacent judging bays, nor the cost judging team. Teams may swap out team members as required to best complete the cost event. There is no allocated structure for how a team presents, and it is the expectation of the cost team that any resources or persons needed for the cost judging are present as the team requires them.

PRESENTATION EVENT

Event Captain: Vincent Chu

Dates/Times: Refer to page 19

The Presentation Event assesses the ability of teams to create a business case for their vehicles to demonstrate their fit within the motorsport landscape of Australia.

The presentation should address the marketability, manufacturing feasibility and profitability of their design, and include an understanding of the marketplace and target audience.

Teams will present themselves in the judging area at the time designated in the static events schedule. Each team will be assigned a 40-minute slot that includes time for equipment setup, presentation, Q&A deliberations between judges and feedback to the teams.

Concept and Deliverables

Refer to the Presentation Event Concept published on the FSAE-A website for outline of the Concept, Deliverables and Sample Judging Questions - Link: [Business Presentation Concept 2025 Document](#)

- Refer to the Presentation Event webinar available at <https://www.saea.com.au/formula-saea-webinars>
- Additional information on what makes a winning presentation will be available at <https://www.saea.com.au/rules-documents-templates> no less than one month before the event.

Equipment

- While there is a projector available, teams are required to bring their own equipment including any cables, adapters, USB sticks and laptops required to run their presentations.

Tips for a good presentation:

- Ensure your presentation does not exceed the allocated time.
- Spell check all documents, and dress appropriately for the event.
- Know your material - practise your presentation beforehand and avoid reading directly from note cards and from the presentation itself.
- Consider alternative options if equipment fails, e.g., have a spare laptop.
- Ask someone from your team to video the presentation and the judges' comments.
- Select presenters who are charismatic and good at public speaking.

Addendum

Teams are reminded of the Addendum for FSAE-A which adds the following additional clauses:

Clause S.2.3: At the Australasian Event, teams are allocated a period of 10 minutes for the team's initial presentation. Teams should try to operate within this allocation, and Points will be deducted for significant variations as follows:

- A presentation duration exceeding 10 minutes will incur a penalty of 2 points;
- A presentation duration exceeding 12 minutes will incur a penalty of 4 points, and will be stopped by the judges; and
- Any content presented outside the allocated 10 minute period will not be considered for judging. Note the Q&A portion of the event is designed only to judge the team's ability to respond to questions and does not represent an opportunity to provide additional content to supplement the initial 10 minute presentation.

Clause S.2.10: At the Australasian Event, teams are required to supply a recording of their 10 minute presentation to FSAE-A via the [FSAE-A 2025 Document Submissions Google Form](#), within 2 hours following the completion of their allocated timeslot in the Event Schedule. Non-compliance will incur a penalty of 2 points. FSAE-A will:

- Share these recordings amongst Business Presentation judges only to support standardisation of scoring; and
- Upload the top three scoring presentations to the FSAE-A website following the competition to support transparency.

Clause S.2.9: Teams which do not meet the concept as required by S.2.2.3, may be penalised up to 50% of the score available. The initial scoring rubric may be marked to a maximum score other than 75 points. The final scores will be adjusted so that the highest scoring team(s) will be awarded 75 points, and all other scores will be adjusted in the same ratio, according to the formula:

Final Score = $SR(\text{your}) \times 75 / SR(\text{max})$,

where SR (your) is your team's initial score from the rubric and SR (max) is the score of the team(s) achieving the highest initial score from the rubric.

Judging

A group of up to four judges will score each team based on the Business Presentation Scoring Rubric, which can be reviewed ([here](#)) ([link](#)). This scoring rubric breaks the scoring down into five categories: content, organisation, delivery & visual aids, and questions. The combined scores of the judges may be adjusted to account for variations between different judging groups as part of the scoring standardisation process. There will be up to four streams of judges at this year's event. The judges will rotate through the various streams throughout the day. The judges will also convene periodically throughout the day to discuss the presentations they have seen so far and discuss any outliers in the scoring. This system has been developed to ensure fairness in judging and to help eliminate bias.

Each judging panel will have a stream leader who will collate scores and feedback and run the session.

DYNAMIC EVENTS OVERVIEW

DYNAMIC EVENT REQUIREMENTS

- A minimum of four drivers are required by each team to compete in all dynamic events.
- Each driver may compete in a maximum of three dynamic events.
- Any individual driver cannot compete in both heats of any dynamic event.
- Endurance/Efficiency counts as two events with the two events run simultaneously.

GENERAL PROCEDURES FOR ALL DYNAMIC EVENTS - STARTING AREAS AND GATES

- For all Dynamic Events at the Australasian Competition, a physical “Gate” will apply in the Staging area.
- The running times for all events as listed in the Handbook may be modified on the actual day subject to circumstances. The latest timing will be announced/displayed at the start of each day.
- Teams should present for each event as soon as ready and/or in accordance with any advised schedule. When ready to run they should enter the Staging Area and will be classified as “Inside the Gate”. No work other than adjustments not requiring tools may be performed on the vehicle inside the gate. If any work is subsequently required, the vehicle must be taken outside the gate.
- Closing of the event will proceed via announcements that the gate is closing 15 minutes before closure and at 5 minutes before closure. It is the responsibility of teams to ensure they are aware of these times.
- All vehicles inside the Gate and capable of running at the time of closure will be allowed to compete. Vehicles not inside the Gate in a ready to run condition at the Closure time will forfeit their right to compete or undertake any further runs in that event.

YELLOW FLAG

- The driver is too slow to a walking pace (10km/h) in the yellow flag zone and may go off course to avoid the stopped vehicle and recovery team without incurring a penalty.
- A driver may be directed by a member of the recovery team to go off course to get around the stopped vehicle.

RED FLAG

- The driver must stop immediately on the track and remain in the vehicle, except in the case of a Vehicle Fire. The engine may be stopped if deemed necessary by the driver.
- The driver will be directed by a person approaching and talking to them when it is safe to return to the pit area at a walking pace.

BLACK FLAG

- The driver must immediately enter the pits on the lap that you are shown the black flag. Unlike other forms of motorsport, you do not have three laps to serve the black flag.
- A driver will be given a black flag for either bad driving or an unsafe vehicle.
- A rolled black flag pointed at the driver will be an early indication of driving that may result in a black flag.

In cases of excessive non-compliance with safety related flag directions, specific penalties will be applied to teams whose drivers are assessed as not complying with flag directions. These will be assessed by the officials but if a penalty is assessed as being required, the penalties will generally range from 5 to 40 seconds and up to disqualification for major safety/Black Flag infringements.

Additional Post Endurance penalties may be applied post-event as covered under Clauses D.14.2 and D.14.3. **Refer to the Addendum.**

DYNAMIC EVENTS OVERVIEW

PENALTIES FOR FLAG VIOLATIONS

In cases of excessive non-compliance with safety related flag directions, specific penalties will be applied to teams whose drivers are identified as not complying with flag directions. These will be assessed by the officials but if a penalty is assessed as being required, the following penalties will generally be applied:

- Failing to pull into the passing diversion under a Blue Flag direction – 5 Second penalty per flag point and/or up to a Black Flag.
- Failing to slow down under a Yellow Flag direction – 10 seconds per flag point.
- Failing to Stop under a Red Flag direction – 40 seconds per flag point and/or disqualification.
- Failing to obey a Black Flag – will be assessed on number of non-compliances but may be held in the driver change area and could result in added time penalties and may include disqualification.

VEHICLE FAILURE OR STOPPAGE

- If your vehicle fails/stops on the circuit, you are to remain in it until the recovery crew arrives.
- The driver has up to two minutes from the vehicle stopping to restart the engine/electric motor(s) before the vehicle will be removed from the circuit.
- If you are pushed from the circuit, please steer the vehicle to the centre of the track unless directed otherwise by the recovery crew.
- Try to get the vehicle into neutral or hold in the clutch to assist the recovery crew to push you from the track.
- The steering wheel should be left with the vehicle.
- Once out of the vehicle, remove your helmet, gloves and open your clothing to reduce heat stress.

VEHICLE FIRES

- If your vehicle is on fire, exit immediately and move away towards the centre of the track.
- A vehicle fire will result in a red flag.

CLARIFICATION ON TYRES, COMPOUNDS AND CHANGES

1. Teams should have two sets of tyres - a Dry set and a Wet set.

2. Within each vehicle set, the tyres do not have to be identical composition or size;

3. Teams can have extra Dry or Wet tyres, but they must not introduce any difference to the basic sets of tyres as approved at Technical Inspection; a new compound or size cannot be introduced on any spare tyres that is put onto the vehicle. The replacement tyre must be identical to the tyre it is replacing on the vehicle at that particular corner of the vehicle.

4. If there is a difference in tyres within a TI approved set of Dry (or Wet) tyres, then, if replaced, each tyre must maintain that same combination as presented at TI.

5. So, if the following exists at TI, then any front tyre must be replaced with an A and a rear tyre with B; an A tyre cannot replace a B tyre or vice-versa.

Vehicle Position	Front LH	Front RH	Rear LH	Rear RH
Tyre Type	A	A	B	B

PRACTICE TRACK

Event Captain:

Dates/Times: Refer to page 19

- Teams will be allowed to practice on a first come, first served basis. The driver has a maximum of five minutes to perform test manoeuvres within the track limits.
- Vehicles will not be permitted to enter the Practice Track unless all inspection stickers are displayed on the vehicle, or the team has been given dispensation by the Clerk of Course.
- Operation of the Practice Track in damp conditions is at the discretion of the Clerk of Course.
- If the vehicle sustains any damage, the TI sticker will be removed.

ACCELERATION EVENT

Event Captain:

Dates/Times: Refer to page 19

- Event will be held on the track adjacent to the drag strip. Vehicles will double back using the dedicated lane defined by cones next to this track.
- A maximum of two drivers can compete in the event, with a maximum of two-timed runs per driver.
- Run order will be based on a randomised order, published before the event. There will be a start time for Driver 1 and Driver 2, split equally across the event open time. Teams that are not ready-to-run when scheduled may be penalised.
- After the first timed run, the driver has the option of taking a second timed run or leaving the staging area. Driver changes are not permitted inside the staging area.
- Gate closure will occur 30 minutes before the listed event closure. Teams must be lined up by 30 minutes before the event closure to complete one final timed run but may forfeit this if they are not ready-to-run within 30 seconds of being directed, as judged by the Clerk of Course. Discretion may be exercised for teams that have completed two timed runs or less.
- No work may be performed on vehicles inside the staging area. Teams must exit the staging area, perform the work, and then return to the back of the line.
- A two-second penalty will be assessed per cone knocked down or out of position.

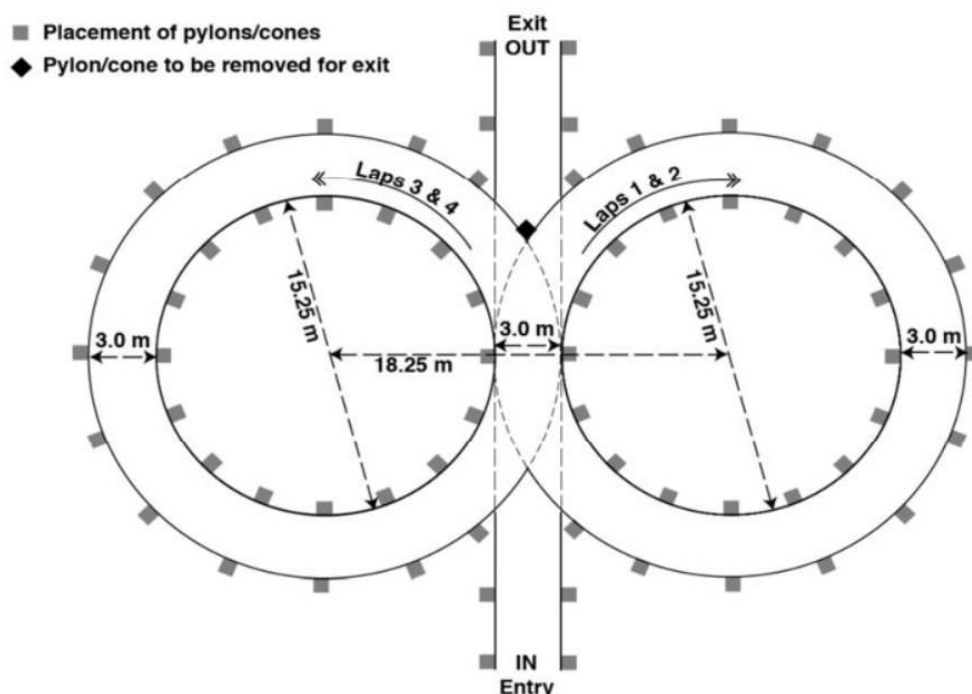
SKID PAD EVENT

Event Captain:

Dates/Times: Refer to page 19

A maximum of two drivers can compete in the event, with a maximum of two-timed runs per driver.

- Run order will be based on a randomised order, published before the event. There will be a start time for Driver 1 and Driver 2, split equally across the event open time. Teams that are not ready-to-run when scheduled may be penalised.
- After the first timed run, the driver has the option of taking a second timed run or leaving the staging area. At the completion of the driver's second timed run, the vehicle must exit the staging area and return to the back of the line. Driver changes are not permitted inside the staging area.
- Gate closure will occur at 30 minutes before the listed event closure. Teams must be lined up by 30 minutes before the event closure to complete one final timed run but may forfeit this if they are not ready-to-run within 30 seconds of being directed, as judged by the Clerk of Course. Discretion may be exercised for teams that have completed two timed runs or less.
- No work may be performed on vehicles inside the staging area. Teams must exit the staging area, perform the work, and then return to the back of the line.
- A two second penalty will be assessed per cone knocked down or out of position.



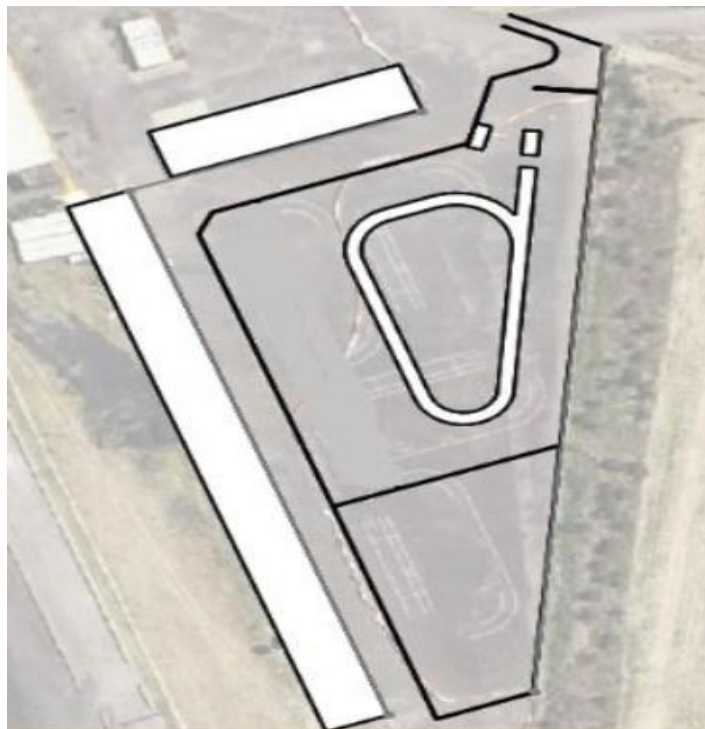
AUTONOMUS VEHICLE DEMONSTRATION

Event Captain: Nick Owen

Dates/Times: Refer to page 19

- The 2025 AV demonstration will consist of scrutineering including EBS test, AV design lite and dynamic track drive.
- Teams wishing to participate in the AV demonstration should note that all vehicles will need to comply with FSAE-A general rules, Formula Student AV rules and the Local Addendums. AVs participating in the demonstration event will need to progress through all scrutineering stages as detailed in the rules, including an autonomous EBS Test (page 25).
- As this year will be a Demonstration Event only, any teams who feel they cannot comply with any part of these rules will need to submit a Rules Enquiry, in the normal way, ahead of the event so that officials can consider what, if any, discretion can be given to that team, noting that compromising safety will not be accepted. The Vehicle Status Video (VSV) is required and submission of the VSV shall be made no later than 2 weeks prior to the commencement of the 2025 FSAE-A competition. The video shall content sequences detailed in FS Rules A5.6.2.
- Teams must supply their own CAN data logger cables and ensure that the cable is fitted to their vehicles prior to AV scrutineering. Data logging equipment will be connected during AV scrutineering and will remain with the vehicle until the last dynamic AV event.
- No work may be performed on vehicles inside the staging area. Teams must exit the staging area, perform the work, and then return to the back of the line.

(Below picture is a representation only and may change)



AUTOCROSS EVENT

Event Captain: TBA

Dates/Times: Refer to page 19

- The Autocross event will be held on a southern loop of the main race circuit, using a shortened version of the same course layout as the Endurance Event. Four or more vehicles may be on circuit at any given time.
- A maximum of two drivers can compete in the event, with a maximum of two-timed runs per driver.
- Vehicles will run from fastest to slowest based on the Skid Pad Event times set.
- Any vehicle that stops on track during the event will not be recovered to the pit area. The recovery crew will instead move the disabled vehicle to a safe location adjacent to the circuit. The Clerk of Course may authorise vehicle recovery during an extended period of stoppage.
- Yellow or red flags will be used to address any circumstances which affect the safe running of the event. Vehicles which are completing timed runs and have been adversely affected, as judged by the Clerk of Course, may be given the opportunity to re-run their timed lap.
- Vehicles that are not ready to commence a timed run within 30 seconds of being directed, as judged by the Clerk of Course, may be asked to exit the staging area, and return to the back of the line.
- No work may be performed on vehicles inside the staging area. Teams must exit the staging area, perform the work, and then return to the back of the line.
- A 2 second penalty will be assessed per cone knocked down or out of position. A 20 second penalty will be assessed for going off course and not re-entering at a point prior to the missed gate. Missing one or more gates of a given slalom counts as a single off course penalty. All cones can be scored as penalties, including those before the start line and after the finish line.

ENDURANCE & EFFICIENCY EVENTS

Event Captain: TBA

Dates/Times: Refer to page 19

- The event will be held on a southern loop of the main race circuit using an extended version of the same course layout as the Autocross Event. Four or more vehicles may be on the circuit at any given time.
- Two drivers must compete in the event, with a driver change taking place mid-event.
- Vehicles will run from fastest to slowest based on the Skid Pad Event times set.
- Teams that are not ready to run when scheduled will be penalised two minutes and may then run at the discretion of the Clerk of Course. Discretion may be exercised for EV class teams who are running out-of-order due to charging limitations. Teams are solely responsible for being ready to run when scheduled.
- Gate closure will occur 30 minutes before the listed event closure, with a hard event closure at the listed time. Teams must be lined up 30 minutes before the event closure to compete in the event but may forfeit this if they are not ready to run within 30 seconds of being directed. The event closure time may only be extended if exceptional circumstances exist, at the discretion of the Clerk of Course.
- Vehicles that are not ready to commence a timed run within 30 seconds of being directed, as judged by the Clerk of Course, may incur a time penalty.
- Any vehicle that stops on track during the event will not be recovered to the pit area. The recovery crew will instead move the disabled vehicle to a safe location adjacent to the circuit. The Clerk of Course may authorise vehicle recovery during an extended period of stoppage.
- No work may be performed on vehicles inside the staging area. Teams must exit the staging area, perform the work, and then return to the back of the line.
- A 2 second penalty will be assessed per cone knocked down or out of position. A 20 second penalty will be assessed for going off course and not re-entering at a point prior to the missed gate. Missing one or more gates of a given slalom counts as a single off-course penalty. All cones can be scored as penalties, including those before the start line and after the finish line.

EFFICIENCY FOR IC CLASS: Efficiency will be calculated based on the energy of the fuel consumed. The vehicle will start the Endurance Event fuelled to the 'full' mark and will be refuelled to the same mark after being released from Parc Ferme.

EFFICIENCY FOR EV CLASS: Electrical energy consumption will be measured by an energy meter supplied by the organisers. The electrical energy consumption and accumulator temperature data will be downloaded by an official during Parc Ferme.

PARC FERME

Event Captains: TBA

Dates/Times: Refer to page 19

Preamble: All vehicles must present to the Parc Ferme area at the end of their Endurance Event whether they finish the event or not. This provides an opportunity for the committee to retrieve track Transponders, Energy Meters and iButtons from each vehicle, and for officials to confirm each vehicle competed in the same form as was presented in the Design Event and at Technical Inspection.

The Clerk of Course or TI Captains may also require any vehicle to attend Parc Ferme following any other dynamic event at their discretion.

Entry: Before entering Parc Ferme, teams may not work on their vehicle, change its settings, or communicate with it in any way (including wireless transmission). To enter Parc Ferme, after either successfully crossing the finish line, or after the vehicle is retrieved from the track at the end of the Endurance event, drivers will:

- Drive their vehicle into the track marshalling area at walking pace.
- Shut down the vehicle in a safe place and activate the kill switch.
- Remain in the vehicle until their vehicle is escorted by an official to the Parc Ferme area.

Process: While in Park Ferme, vehicles will complete the following processes:

- TI at the discretion of the Parc Ferme Captains.
- Track transponders will be removed by officials.
- For EV vehicles, teams will remove their Energy Meter systems (EM box and CT box) from their vehicles under the supervision of an official.
- For EV vehicles, teams will disassemble their car as much as is required to allow the accumulator thermal strip to be viewed by a parc ferme official.
- IC vehicles that have successfully completed their Endurance event will be escorted to the refuelling station and refuelled under the direction of an official. These cars will then be escorted back to the Parc Ferme area for any further inspection that may be required.
- If any issues are identified during the Parc Ferme process, the vehicle will be impounded for detailed inspection and/or teardown.

Exit: Once all Parc Ferme processes are completed, and signed off by an official, teams will return their released car to their pit. Note that a team's results for all events will remain provisional until their vehicle has been released from Parc Ferme.

PENALTIES: A vehicle failing Parc Ferme will be subject to penalties up to and including disqualification from all dynamic events.

- Vehicles may be inspected at random or at the specific interest or concern of officials.
- Any reasonable measuring devices may be used in accordance with the practice at TI, Noise Test and/or Fuel Station.
- Any aspect of the vehicle may be checked against the rules, regardless of whether the aspect is included on the TI schedule or not.

TI does not cover all areas of the rules, nor the full detail of all rules. However, the onus is always on the competing teams to ensure that their vehicle is in complete compliance to the rules, always during the event.

PARC FERME (continued)

The following process will be applied for assessment of non-rules compliant conditions:

- 1) A non-compliant condition (NCC) is identified on a vehicle.
- 2) Is the NCC an item which was checked at TI? (**Yes = Code TY, No = Code TN**)
- 3) The NCC probably occurred because of deliberate action rather than just due to vehicle operation without deliberate action (**Yes = Code DY, No = Code DN**)
- 4) The NCC is an item which could affect vehicle performance? (**Yes = Code PY, No = Code PN**)
- 5) The NCC is an item which could affect vehicle safety? (**Yes = Code SY, No = Code SN**)

INDICATIVE PENALTY MATRIX

	TY		TN	
	DY	DN	DY	DN
PY	Red	Yellow	Orange	Salmon
PN	Salmon	Green	Yellow	Blue
SY	Red	Yellow	Orange	Salmon
SN	Salmon	Green	Yellow	Blue

HIERACHY OF PENALTIES



RED: Disqualification.

ORANGE: Even though not measured at TI, for apparently deliberate NCC's which affect performance or safety, it is appropriate to apply a significant penalty as vehicles must be within the rules, regardless of whether it is something that was checked at TI. The greater the level of variation, the greater the penalty that will be applied. For small variations in a dimensional item, a base penalty of 50 points will be applied. For more significant deviations from the rules, additional penalties will be applied. Additional penalties of either 50 points or 75 points will apply, dependent on the NCC being classified as minor or major by the officials.

YELLOW: For apparently non-deliberate NCC's but which do affect performance or safety, it is appropriate to apply a meaningful penalty as vehicles must be maintained within the rules, and as checked at TI. The greater the level of variation, the greater the penalty that should apply. Very small variations to a measurement at TI, this will be ignored as an NCC if the officials regard it as within reasonable measurement and repeatability errors. A minimum penalty of either 75 points or 100 points will apply, dependent on the NCC being classified as minor or major.

SALMON: This code covers the group of actions whereby deliberate actions have been taken to create a variation to the rules for some reason, but there is no apparent performance advantage or adverse safety implications. This also covers items which may affect performance or safety, but were not identified prior to Parc Ferme, and which appear not to have arisen from deliberate action. Because of the resultant potential advantageous effect, and non-compliance with the rules and/or safety during the event, a team will receive a flat penalty of 50 points.

GREEN: Items checked at TI, and which have changed with usage or repairs or operation of the vehicle, but no performance or safety effect results. However, teams should ensure their vehicles remain fully compliant with what was checked at TI, and a flat penalty of 25 points will be applied.

BLUE: Will result in no first-time penalty for the team. However, if a similar NCC occurs in the next year's event, then a penalty will be applied.

COFFEES

The following Cafés are open early, so you can get your coffee before the action starts. Coffee can also be purchased onsite.

McDonalds Southbound

Lot 2 Thompsons Road, Keilor North: Open from 5:00 - 22:00 Daily.

McDonalds Northbound

BP Service Centre, Calder Hwy, Keilor North: Open 5:00 - 22:00 Daily.

LOCAL SHOPPING

Watergardens Town Centre Stores:

- Aldi
- Coles
- Woolworths
- Pharmacies
- Banks
- Specialty shops
- Eateries

ONSITE CATERING

Calder Park Raceway - Speed Cafe will be open during the event.

EQUIPMENT SUPPLIERS

Repco

8/460 Melton Highway, Taylors Lakes

Ph: (03) 8390 8044

Opening Hours:

Monday to Friday: 8am-5pm

Saturday: 8am-4pm

Sunday: 9am-3pm



2/85 Vineyard Road, Sunbury

Ph: (03) 9744 7288

Opening Hours:

Monday to Friday: 8am-5pm

Saturday: 8am-4pm

Sunday: 9am-3pm

Bunnings

430 Melton Highway, Taylors Lakes

Ph: (03) 9449 6000

Opening Hours:

Monday to Friday: 6:30am- 9pm

Saturday/Sunday: 7am-7pm



95 Vineyard Road, Sunbury

Ph: (03) 9744 0600

Tuesday to Friday: 6:30am- 9pm

Saturday to Monday: 7am-7pm

SuperCheap Auto

430 Melton Highway, Taylors Lakes

Ph: (03) 9390 9699

Opening Hours:

Monday to Thursday: 8:30am – 5:30pm

Friday: 8:30am- 7pm

Saturday: 8:30am- 5pm

Sunday: 9am-5pm



49-51 Horne Street, Sunbury

Ph: (03) 9746 3610

Opening Hours:

Monday to Friday: 8:30am – 5:30pm

Saturday: 8:30am- 5pm

Sunday: 9am-5pm

VEHICLE SHIPPING

These procedures serve as a starting point for international teams shipping vehicles to Australia. Please consult with your shipper or freight forwarder to ensure your shipment meets their specific requirements. You should also ensure your shipment complies with all relevant Australian Customs regulations.

Shipping any car may be considered hazardous by the Federal Aviation Administration (FAA). The shipment may be inspected and must be signed off by someone trained in HazMat shipping. Please make sure you fully inform your shipper/freight forwarder of the contents of your shipment to avoid potential complications.

GENERAL RECOMMENDATIONS: Ship early to ensure your vehicle arrives on time for the competition and use a reputable agent who has experience in moving such freight. We recommend you also engage someone on-the-ground in Australia who can handle the customs clearance and serve as a point of contact. SAE-Australasia staff cannot provide advice on shipping, customs regulations or visas.

INBOUND SHIPMENTS: All labels and documents associated with the shipment must show the team or university as the receiving party. Do not ship your vehicle with either the organiser or competition site listed as the receiving party. Neither SAE-Australasia or Calder Park Raceway will execute customs or shipping documents of any type.

Use the following address for all inbound shipments:

[Insert name of your university] c/- Calder Park Raceway
377 Calder Freeway Calder Park, Victoria Australia 3037

Please provide SAE-Australasia (formulasae@sae-a.com.au) with the details of your shipment at least one (1) week in advance so we can advise Calder Park Raceway.

CUSTOMS AND SHIPPING PAPERWORK: All paperwork, documentation and/or forms required for inbound/outbound shipping or customs clearance must be completed and supplied by the school/university. SAE-Australasia or Calder Park Raceway staff cannot prepare any shipping documents for inbound or outbound shipments. All paperwork is the sole responsibility of the team.

CUSTOMS REGULATIONS: It is the responsibility of the team to adhere to all Australian Customs regulations. All inbound shipments are subject to Australian Customs inspections.

TEAM CONTACT: To provide support services for inbound/outbound shipments, and especially to resolve any problems that might arise, each team must provide the names and mobile phone numbers of the Team Captain and the Faculty Advisor.

DAMAGED SHIPMENTS: Inspecting shipments and reporting and documenting damage to the shipment is the sole responsibility of the receiving team. Neither SAE-Australasia or Calder Park Raceway is responsible for damage to your shipment from any cause, including negligence.

VEHICLE SHIPPING (continued)

SHIPPING CONTAINERS: Shipping crates/containers must have hi-low forklift access from ends and sides. The shipping crate/container must fully enclose the car. The base of the container must be strong enough to support the entire weight of the shipment when it is being picked up, rotated and moved by a fork-lift or similar equipment. Be aware that containers are likely to be stored outside and should be weatherproof. Wood containers must be of insect resistant or treated materials suitable for international transportation. Wood crates, containers and pallets should be screwed or bolted together.

SHIPPING CONTAINER MARKING: All shipping containers must have the school's name permanently and clearly marked.

CRATING AND UNCRATING: Crating and uncrating is the sole responsibility of the team. Do not pack the tools you need to open the crate inside the crate itself.

UNLOADING AND LOADING: Loading and unloading is the responsibility of the teams. Calder Park Raceway has a forklift available during business hours for loading/unloading. This should be pre-arranged prior to delivery of the shipment. No team will have access to their vehicle on-site at Calder Park Raceway prior to the Thursday of the event. Teams will however be allowed to unload their vehicle on-site but not unpack prior to Thursday.

OUTBOUND SHIPMENTS: All shipments must be properly packed in the containers and labelled before the team leaves the site on Sunday or Monday. It is each team's responsibility to schedule the pick-up of your outgoing shipment and prepare all the paperwork. All shipping and customs forms must be filled out by team/university representatives. It is a requirement that the return freight is arranged prior to departing from your home city for the event. Calder Park Raceway personnel are not permitted to make arrangements for export shipping and scheduling.

REQUIRED INFORMATION FOR ALL SHIPMENTS: You must email the information, for both inbound and outbound freight, as outlined below to the event organisers (formulasae@sae-a.com.au) at least one week before your shipment is scheduled for delivery or pick-up. Please provide the following information:

Inbound Shipment

- Shipping company.
- Shipping company contact name.
- Phone number and email.
- Scheduled arrival date/time for inbound shipment.
- Shipment/shipper identification number.

Outbound Shipment

- Shipping company.
- Shipping company contact name.
- Phone number and email.
- Scheduled date/time for outbound pick-up.
- Shipment/shipper identification number

VOLUNTEER INFORMATION

Thank you for serving as a volunteer at the Formula SAE-Australasia event. This event is conducted by SAE-Australasia under a Motorsport Australia (MA) permit and the MA OH&S policy. This policy can be viewed on the MA website (motorsport.org.au) or at Administration. You are involved as an official by your own choice, and must bear responsibility for your own welfare, as well as those around you.

The following people serve as key points of contact, and will be happy to assist you:

- **Angela Krepcik:** Volunteer Coordinator and SAE-A General Manager
- **Adrian Feeney:** Event Manager
- **Nathan Tarlinton:** Clerk of Course/ Protest Committee
- **Michael Lambert:** Facilities & Logistics Manager
- **Rose De Amicis:** Event Administration

If in doubt, ask one of the volunteers in an orange safety vest with the title on their back “**EVENT COMMAND**”.

VOLUNTEER SIGN-IN: All volunteers must visit Administration to sign-in upon arrival each day.

SAFETY: If you see something unsafe, or if you are uncertain about something, then please bring it to the attention of Administration, or one of the above-mentioned people. Please also ensure you drink plenty of water and apply sunscreen regularly. Water and sunscreen will be available in all event locations. A hat and long sleeve shirt are also recommended. High-visibility vests must be worn at all times while on duty and will be supplied by the event organisers. Event captains will discuss the safety of individual events with their volunteers prior to commencement.

MOTORSPORT ANTI-DOPING POLICY: All officials must comply with the motorsport anti-doping policy. Do not take any drugs (including alcohol) which may affect your attention and judgement. The MA policy states that officials may be subject to random drug and/or breath testing at any time during an event. This regulation serves as the required 24-hour notice. The consumption of alcohol by drivers, crew and officials is forbidden until the conclusion of the event, and spectators are forbidden to take or consume alcohol onsite.

EVENT SCHEDULE: All volunteers must attend their allocated event(s). This information can be found outside Administration. Although you have been allocated a specific role, you may be asked to perform other activities. Your flexibility and cooperation will be greatly appreciated.

MISSED BRIEFINGS: If you miss a briefing, please report to your Event Captain before commencing your duties.

MEALS: Volunteer meals will be served each day in the volunteer area of the cafe. Please refer to your event guide for the meal break times. Please advise the Volunteer Coordinator if you have any special dietary requirements.

VOLUNTEER INFORMATION (continued)

PERSONAL BELONGINGS: We cannot facilitate the storage of bags and valuables. Please bring only what you need and leave other items in your vehicle.

SMOKING: Smoking is only permitted in the designated smoking area.

REPORTING INJURIES: If you are injured or become ill, please report it to your Event Captain immediately. All injuries/illnesses must be reported to a medical officer so that they can be identified, checked, recorded, and appropriate care administered. If at any time you are unable to carry out your duties because of personal limitations or because of ill health, please report to your Event Captain or the Event Manager.

EVENT CHECK-IN: Please use the following guidelines when commencing at your allocated event:

- Check the general area and equipment.
- Report any problems, apparent danger, concerns or shortages to your Event Captain.
- Read and understand all equipment instructions (e.g. fire extinguishers).
- Discuss the duties and experience of each volunteer and their general location during events. Please give thought to the need for all volunteers to be protected by safety barriers and debris fences where appropriate for the dynamic events.
- Discuss the location of equipment, which should be placed where visible and available to others but not in anyone's way.
- Plan escape routes in conjunction with the other volunteers and discuss warnings to be given to each other if necessary (e.g. whistles or hand signals).
- Locate equipment and personal belongings where they will not delay your escape by blocking an escape route or by causing falls.

HOT TRACK GUIDELINES: Your prime consideration must be to protect yourself from danger, then your fellow volunteers, then the drivers and any others in trouble. Do not move in front of barriers unless instructed to do so by a senior volunteer, or unless your duties require you to do so.

- Unless your specific duty requires it (e.g. flag volunteers while flagging) do not station yourself in or behind any gaps in the primary protection barrier, nor in any position less than 1m on either side of any such gaps.
- At all times, stand up, remain alert and face the oncoming traffic unless your specific duty requires otherwise (e.g. yellow flag volunteer). Do not congregate in such a manner that access along the volunteer zone is unduly restricted.
- Do not lean on guardrail, concrete blocks or debris fences - they are designed to move when hit by vehicles.
- No smoking is allowed at any time.
- Be always prepared for fast moving traffic of any nature.
- If an incident occurs in your vicinity, protect yourself from flying debris by turning or moving away or dropping below barriers as time and circumstances permit. Be prepared to use pre-planned escape routes. Warn your fellow volunteers.
- Be aware that running fuel fires can pass under the concrete barriers.

VOLUNTEER INFORMATION (continued)

RIDING ON BACK OF VEHICLES: It is not permitted to ride on the back of any recovery vehicle or other vehicle, or on any trailer, unless this is necessary for the proper performance of their duties, such as the recovery of a vehicle or distribution of food and equipment. Otherwise, all occupants of vehicles must be seated and wearing a seat belt.

OPERATING ON TRACK: Please use the following guidelines when operating on track:

- Check the movement of volunteers and vehicles around you.
- Take care when moving from behind barriers and try to always approach any incident with the oncoming traffic in front of you.
- Be aware that dirt or extinguisher clouds can prevent you from seeing oncoming traffic and also prevent them from seeing you.
- Where possible keep a stopped vehicle between you and oncoming traffic (but not between a stopped vehicle and close-by barrier).
- Do not stand close to any stopped vehicle during competition, even if behind barriers, because of chance of debris if the stopped vehicle is hit by another vehicle.
- Remain exposed for the minimum time necessary to complete your task then get back behind the barrier.

COLD TRACK GUIDELINES: Please use the following guidelines when operating on a cold track:

- Relax but ensure that some volunteers at the post remain alert and watchful.
- Use the time to check equipment and the track.
- Be ready to come to attention immediately if circumstances change.
- Be prepared for traffic to approach from any direction.
- Protect yourself from extremes of weather, sunburn, hearing damage, dust and dirt in your eyes. It is recommended that you wear sunglasses or suitable eye protection. Wear suitable footwear for your duties and gloves if you are likely to be handling hot or sharp material.

Each volunteer must remain responsible for their own welfare. Sunscreen is provided, but please make sure to bring a hat with you.

VOLUNTEER INFORMATION (continued)

Formula SAE-A Volunteers for Static Events Code of Conduct

All volunteers should recognise that we are here to develop and inform students, and help them prepare for real life challenges, should they continue their studies or are about to enter the workforce.

Volunteers should show no discrimination or favouritism towards the quality of vehicles or individual backgrounds of any team member. Naturally, we realise that conversations with teams in their first year of competition may differ from the more experienced teams.

Conduct of Volunteers – our expectations:

1. Officiate in strict accordance with the competition rules including the local addendum.
2. Behave in a positive and courteous way during all interactions with event participants, officials and visitors.
3. Ensure that questions are structured to evaluate the depth and breadth of the students' knowledge and help them expand technically and personally (note that we are not there to catch them out!).
4. If you are a static event judge, only conduct activities witnessed by at least one other event official.
5. Deliver any feedback in an open and constructive manner relevant to all teams (where possible).
6. Only judge teams during the allotted time for that team - any judging activities conducted out of allotted times or outside assigned area of expertise must be avoided, unless requested by the team for clarification and in accordance with points 2, 3, 4 & 5.
7. Take proactive action to avoid any conflicts of interest and declare to the Event Captain if in doubt.
8. Uphold the best interests of the event, fellow officials and organisers, sponsors and participants, both during the event and in public forums beyond the event.
9. Under no circumstances are any volunteers or judges able to communicate results to the teams directly.
10. Actively encourage and train understudies to ensure continuity of the required skill sets.

FORMULA SAE
AUSTRALASIA



***SAE-A APPRECIATES YOUR
INVOLVEMENT AND PARTICIPATION
AT FORMULA SAE-A 2025***

